



ভারতীয় প্রযুক্তিবিদ্যা প্রতিষ্ঠান খড়্গাপুর
भारतीय प्रौद्योगिकी संस्थान खड़गपुर
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

AI Systems

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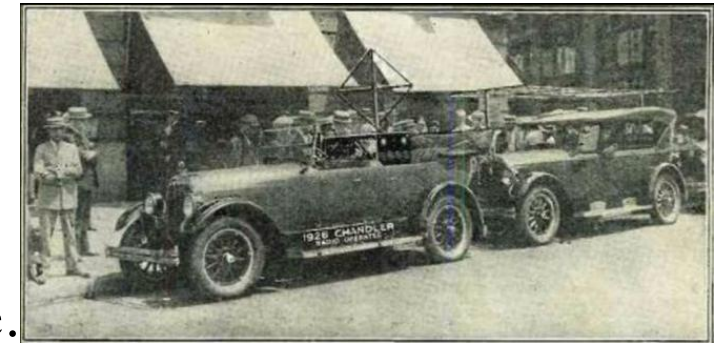
PhD: IIT Indore MTech: IIITDM Jabalpur

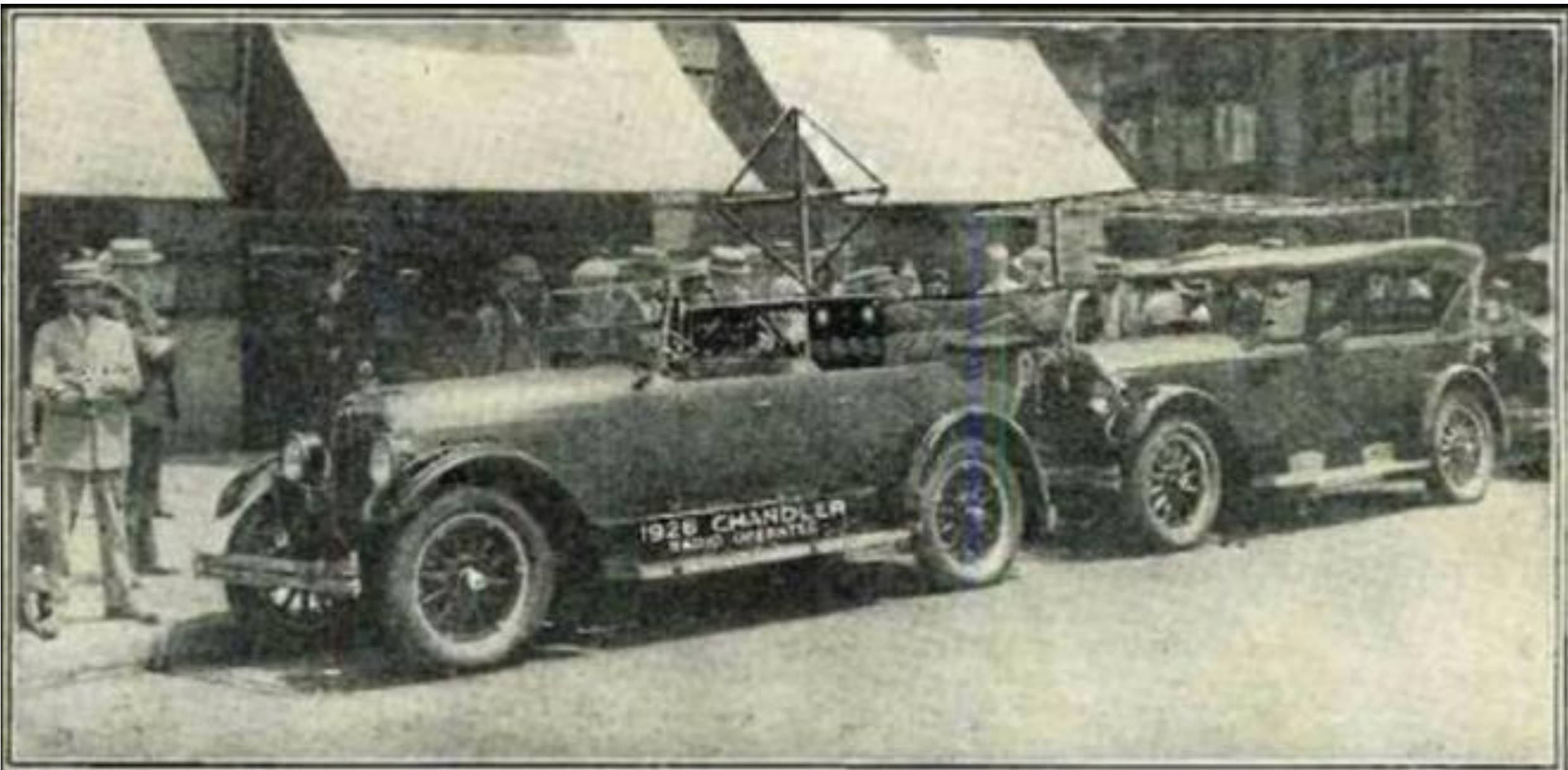


History of Classical AI

1925: Houdina Radio Control

- Francis P Houdina developed a radio-operated automobile.
- He equipped a 1926 Chandler with a [transmitting antenna](#) on the tonneau and operated from a second car that followed it with a [transmitter](#).
- The [radio signals](#) operated small [electric motors](#) that directed every movement of the car.
- In 1925, he publicly demonstrated his [radio-controlled driver-less car](#) American Wonder in New York City streets, traveling up Broadway and down Fifth Avenue through thick traffic

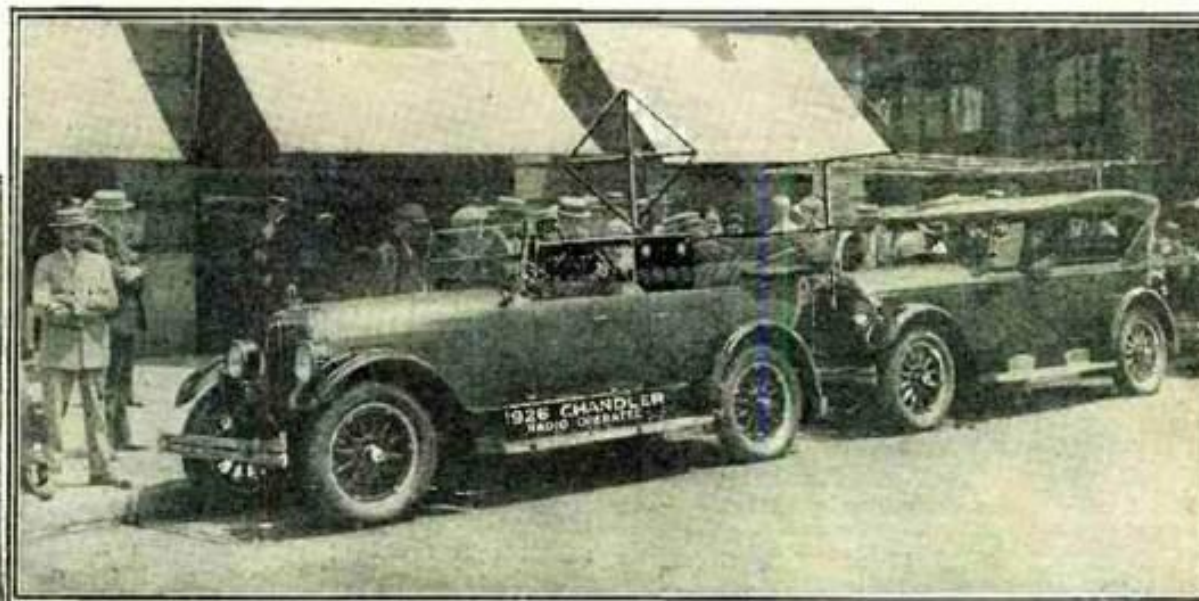
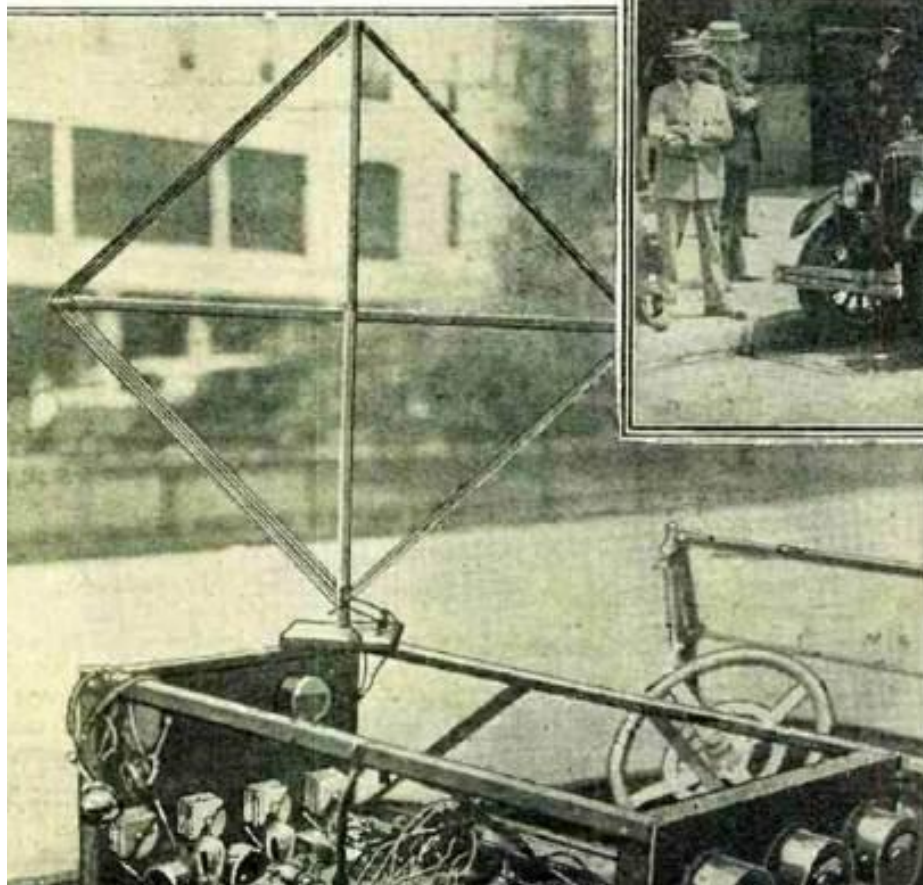




Radio-Controlled Automobile

By HERNDON GREEN

Radio is to control a car in transcontinental tour. The system, which is extremely simple and effective, is fully described here.



THE DIAGRAM

A glance at the wiring diagram given in an accompanying illustration will show the circuit in its simplest form. Some of the amplifying relays are left out in order to simplify the circuits.

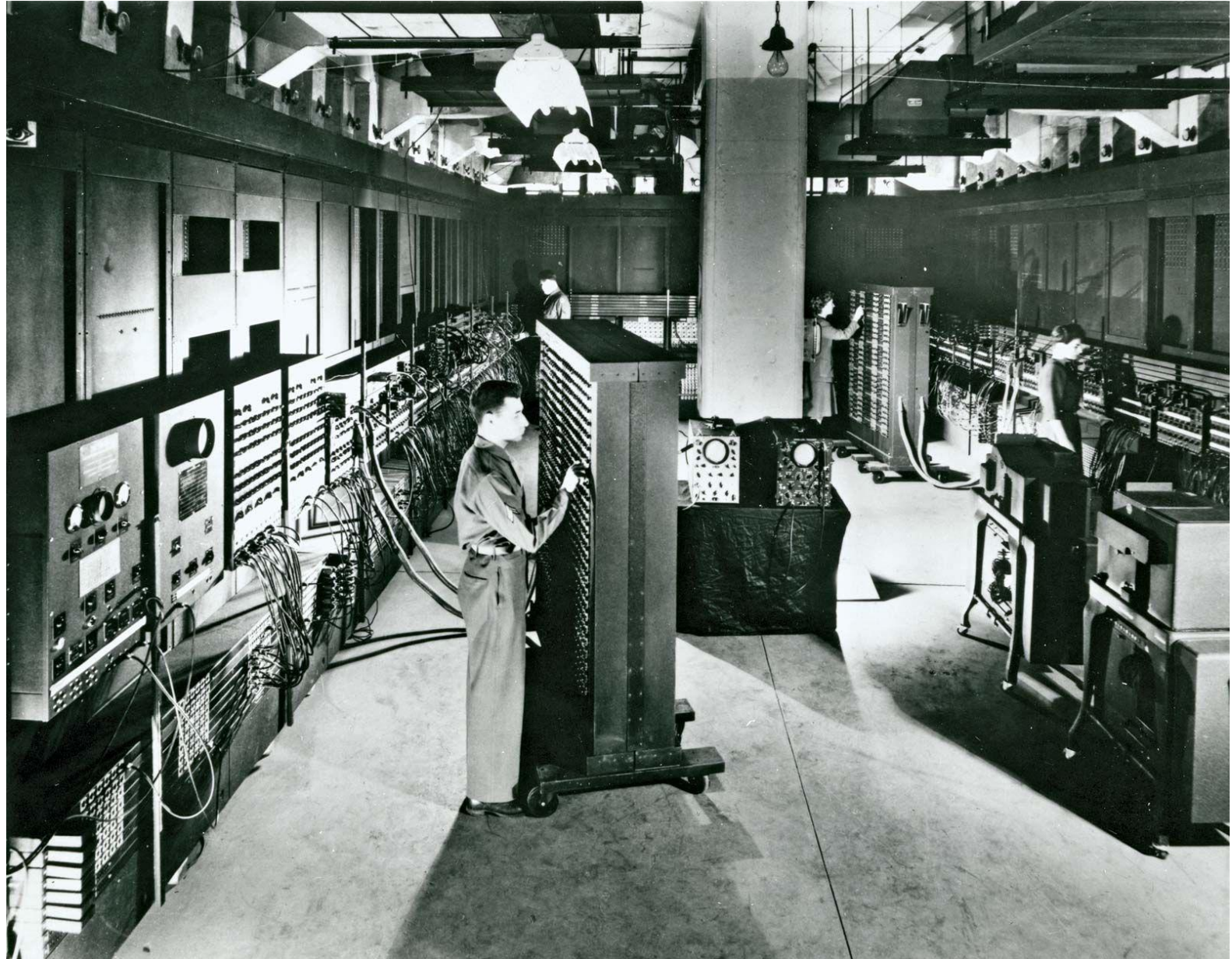
The two transmitters are of the usual 10-watt type, using storage battery supply for the filaments and plates. They are housed in the tonneau of the control car together with the power units. The keys controlling them are placed on a small shelf at the right side of the dash.

The receivers are the usual type, employ two loop and radio frequency coils, and

Above is shown the two cars used in the transcontinental tour by Mr. Francis P. Houdina. The front car has no driver but is controlled by radio from the second car. Note the transmitting antenna and the receiving loop.

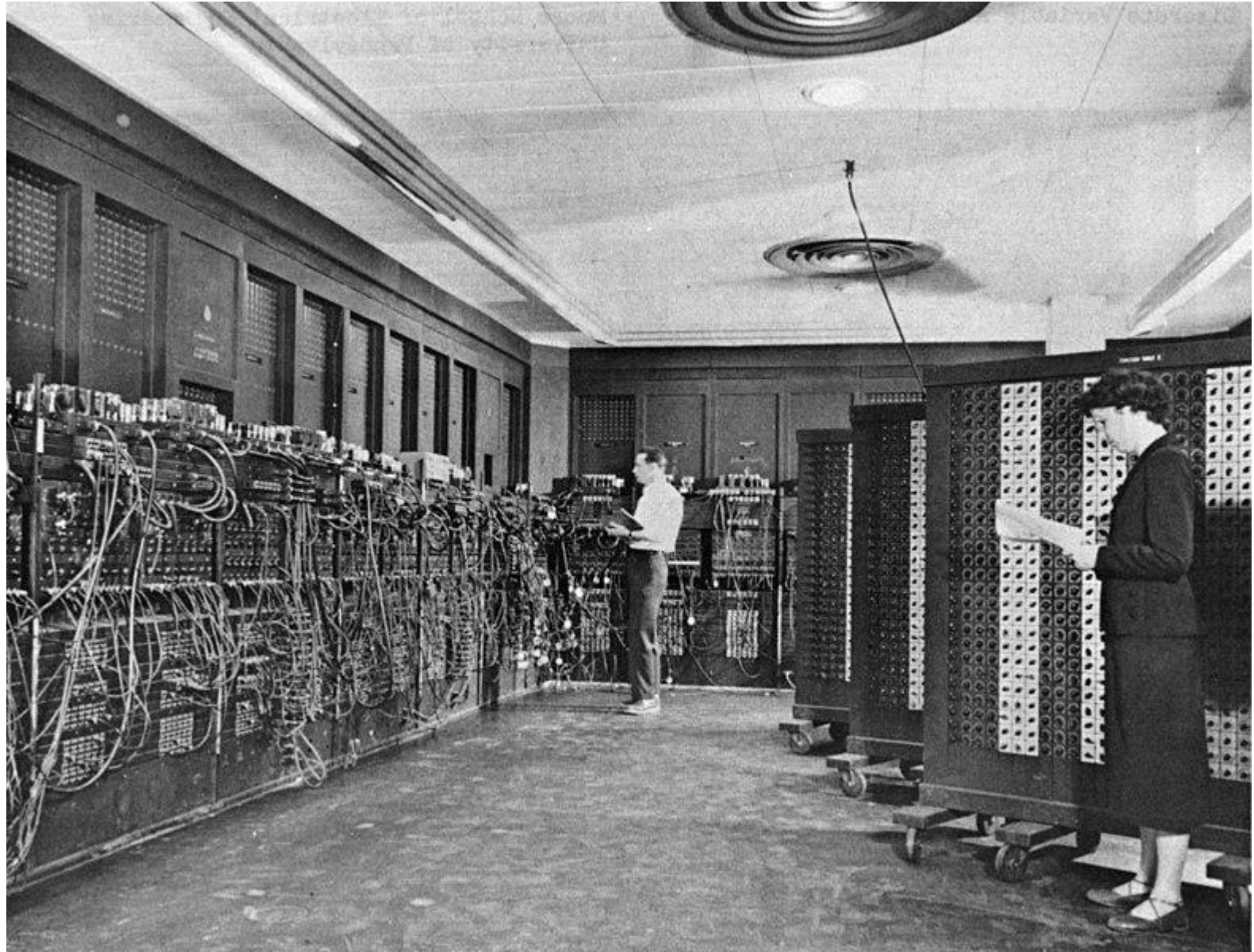
1946-1960

- 1946: ENIAC heralds the dawn of **Computing**



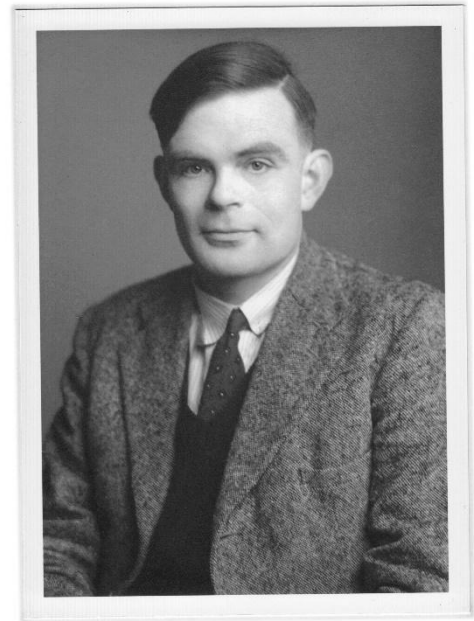
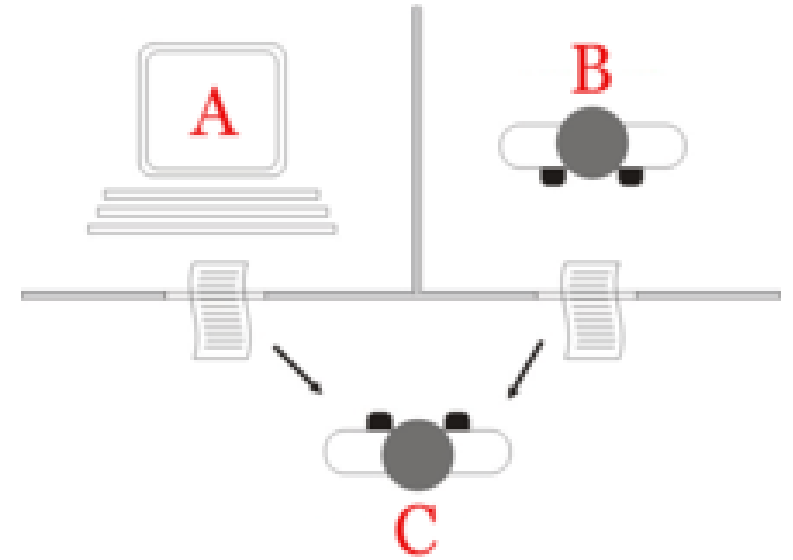
1946-1960

- 1946: ENIAC heralds the dawn of **Computing**



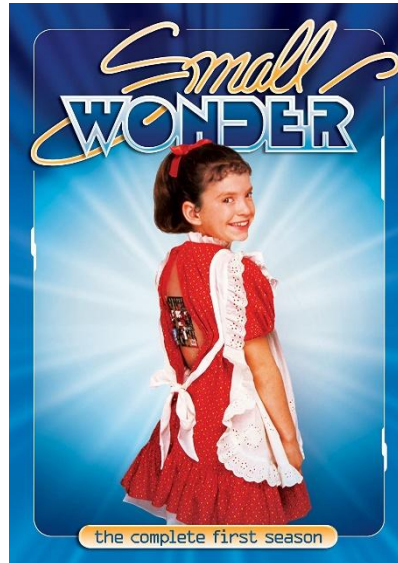
1950: Turing Test

- 1950: Turing asks the question “**Can machines think?**”
- ‘imitation game’ is played with three people,
 - a man (A),
 - a woman (B), and
 - an interrogator (C)
- ‘What will happen when a machine takes the part of A in this game?’ Will the interrogator decide wrongly as often when the game is played like a man and a woman?
- Player C, the interrogator, is given the task to determine which player – A or B – is a computer and which is a human.
- The interrogator is limited to using the responses to written questions to make the determination.



1950: Turing Test

- Turing defined “**intelligent behavior**” as the ability to achieve human-level performance in all cognitive tasks, sufficient to fool an interrogator.”
- Intelligent computer
 - interrogated by a human via invisible communication channel,
 - pass the test
 - “if the interrogator fails to guess there is a computer or a human at the other end”
- Popular Science Fictions
 - Small Wonders
 - Karishma ka Karishma



1956: DARTMOUTH RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

A PROPOSAL FOR THE DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

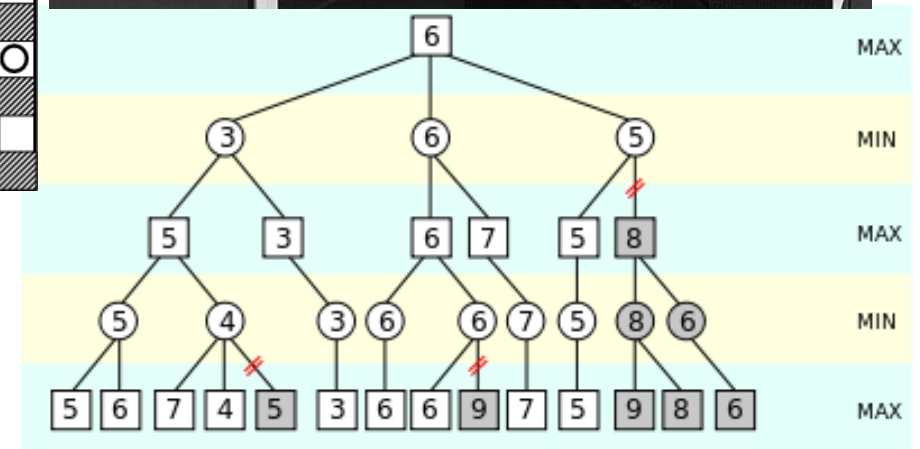
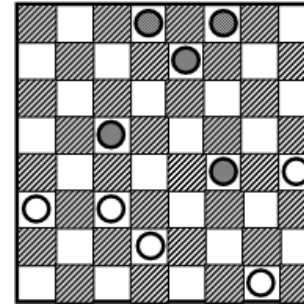
J. McCarthy, Dartmouth College
M. L. Minsky, Harvard University
N. Rochester, I.B.M. Corporation
C.E. Shannon, Bell Telephone Laboratories

August 31, 1955

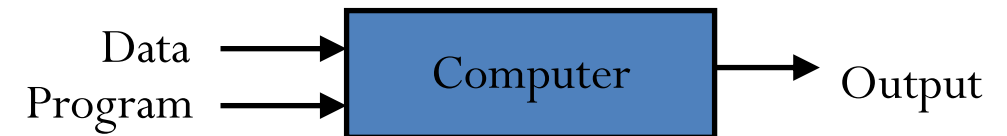
We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.

1959: Machine Learning

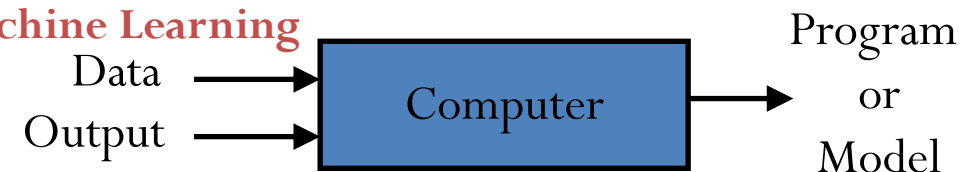
- Arthur Samuel: Automated Game for “Computer checkers” development
- Ability to learn without explicit programming.
 - Samuel implemented **alpha-beta pruning**
 - Grayed-out subtrees don't need to be explored (when moves are evaluated from left to right),
 - the **group of subtrees** yields the value of an equivalent subtree or worse, and
 - as such cannot influence the final result.



Traditional Programming



Machine Learning



1964: Eliza Chatbot

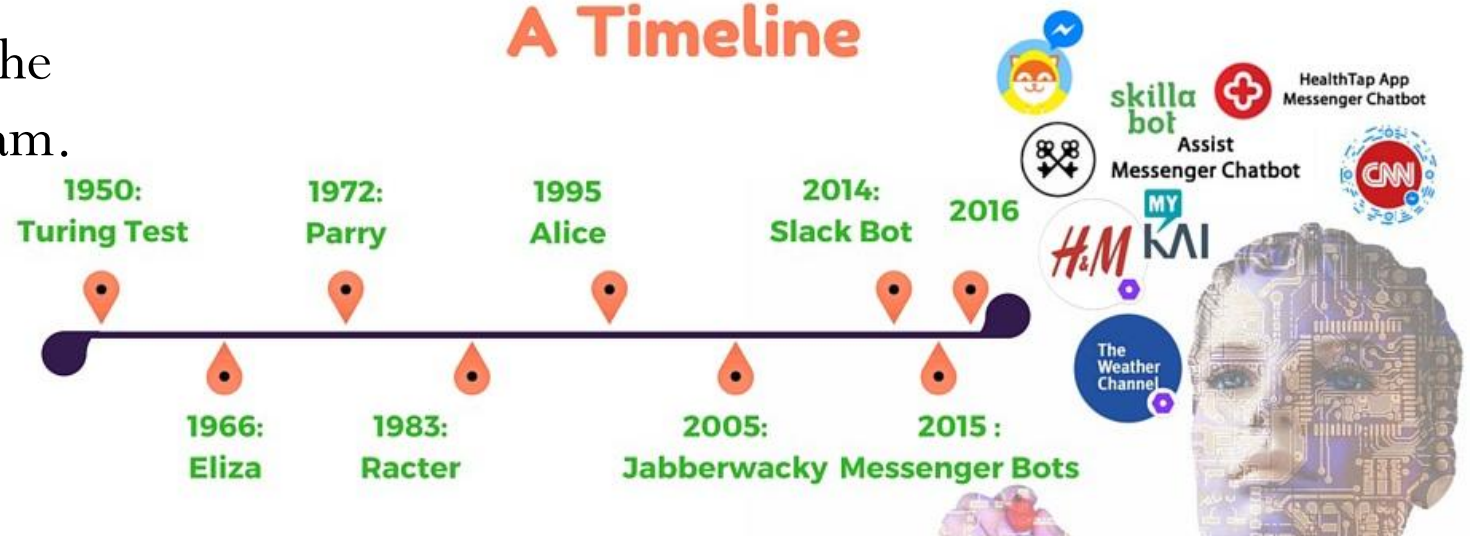
- The chatbot psychotherapist
- Using "pattern matching" and substitution methodology, the program gives canned responses.
- It made early users feel they were talking to someone who understood their input.
- The program was limited by the scripts that were in the program.

Talk to Eliza by typing your questions and answers in the input box.

```
> Hello, I am Eliza. I'll be your therapist today.  
* What is your name?  
> Does that question interest you?  
* Are you intelligent?  
> Why are you interested in whether or not I am intelligent?  
* To do Turing Test.  
> I'm not sure I understand you fully.  
* Do you know Turing Test?  
> Oh... Let's move on to something else for a bit. know Turing Test?
```

TYPE HERE

History of Chatbots: A Timeline



1966: Shakey general purpose mobile robot

- First mobile robot with the ability to perceive and reason about its surroundings
- Tasks required **planning**, **route-finding**, and the rearranging of simple objects.



AI Winter (1974-80)

- Abandonment of AI: There were two major winters in 1974–1980 and 1987–1993
- 1966: failure of [machine translation](#)
- 1970: abandonment of [connectionism](#)
- Period of overlapping trends:
 - 1971–75: [DARPA](#)'s frustration with the [Speech Understanding Research](#) program at [Carnegie Mellon University](#)
 - 1973: large decrease in AI research in the United Kingdom in response to the [Lighthill report](#)
 - 1973–74: DARPA's cutbacks to academic AI research in general
- [1974 – 1980: AI Winter #1](#)
 - Failure of machine translation
 - Negative results in Neural nets
 - Poor speech understanding

AI Winter (1974-80)

- Problem with A.I.
- Applications were primarily for toy problems
- Low computational power
- Combinatorial explosion too many exhaustive Rules
- Common-sense is nearly impossible to program
- Minsky's book = Perceptron showed limitations of simple neural network
- Loss of government funding in A.I.

AI Spring and Summer

AI Spring (1980-87)

- Expert systems used in several real-world applications
- Driverless car
- Hopfield network popularization of backpropagation
- Minsky (1984) – “Winter is coming!”

AI Winter (1987-93)

- 1987 – 1993: AI Winter #2
- 1987: collapse of the [LISP machine](#) market
 - Decline of LISP
- 1988: cancellation of new spending on AI by the [Strategic Computing Initiative](#)
- 1990s: end of the [Fifth Generation computer](#) project's original goals
- 1993: resistance to new [expert systems](#) deployment and maintenance
 - Decline of specialized hardware for expert systems

AI Winter (1987-93)

- Popularity of the P.C.
- Lack of attractive results
- Funding cuts
- Lasting effects
 - “Artificial Intelligence is associated with systems that have all too often failed to live up to their promises.”
 - “Some believe the word 'robotics' actually carries a stigma that hurts a company's chances at funding.”
- Winter over due to
- 1993: Association Rule Mining by Rakesh Agrawal and Srikant Ramakrishnan

AI Spring and Summer

AI Summer (1994-2000)

- **Apriori algorithm: Association Rule Mining** by Rakesh Agarwal in 1993 and 1995
- 1997 **Deep Blue** beats Kasparov in chess
- Theory – Including probability, information theory, optimization etc.
- Moore's Law – Rapid growth of processing power.
 - “Number of transistor doubles every two years”

History of Modern AI

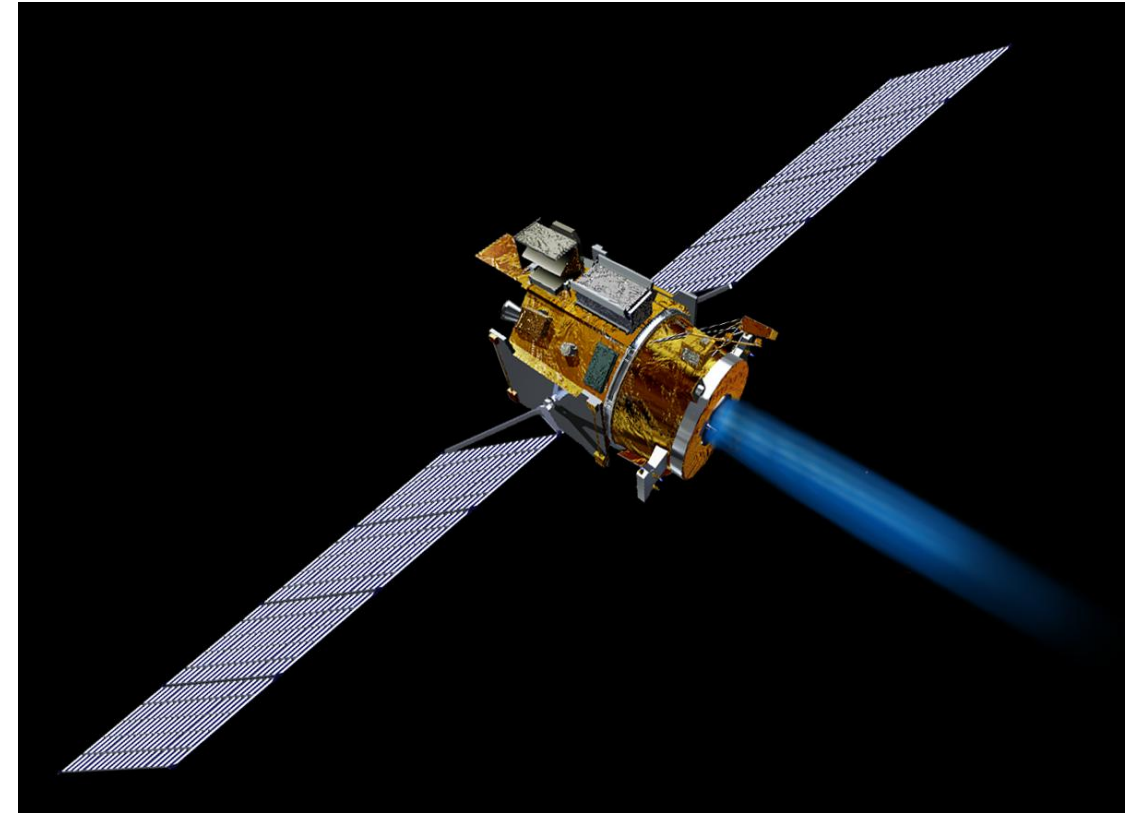
1997: Deep Blue ends Human Supremacy in Chess

- I could feel **human-level intelligence** across the room - Gary Kasparov, World Chess Champion (human)
- In a few years, even a single victory in a long series of games would be the triumph of human genius.
- Does Deep Blue use AI? “If it works, its not AI!”
- “Saying Deep Blue doesn’t really think about chess is like saying an airplane doesn’t really fly because it doesn’t flap its wings”. – Drew McDermott



1998: Deep Space 1

- 1998: Deep Space 1
- For two days in May 1999, an AI program called “Remote Agent” autonomously ran Deep Space 1 (some 60,000,000 miles from earth)
- The spacecraft’s main goal was to test 12 high-risk technologies including
 - ion propulsion,
 - autonomous optical navigation,
 - a solar power concentration array, and
 - a combination miniature camera-imaging spectrometer.



Key Dates

Oct. 24, 1998: Launch

July 29, 1999: DS1 passed by asteroid 9660 Braille

Sept. 18, 1999: Primary mission ended

Sept. 22, 2001: DS1 entered the coma of Comet Borrelly

Dec. 18, 2001: DS1's ion engine is finally turned off

2005 onward: Driverless Cars

- DARPA Grand Challenge
- 2005 Autonomous driving for 135 miles in mountain road of desert
- Stanley and three other cars drive themselves



Now

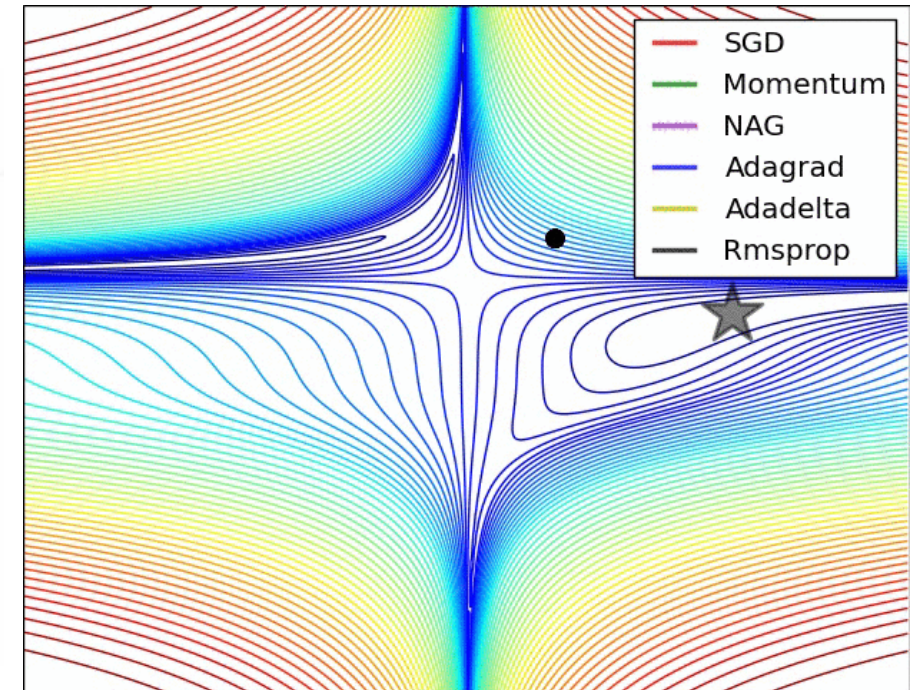
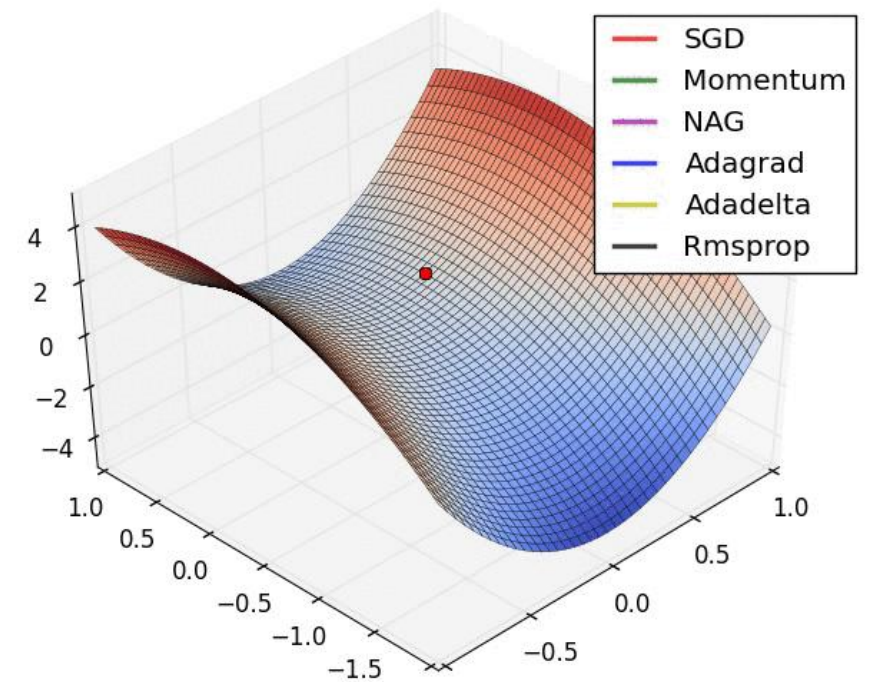
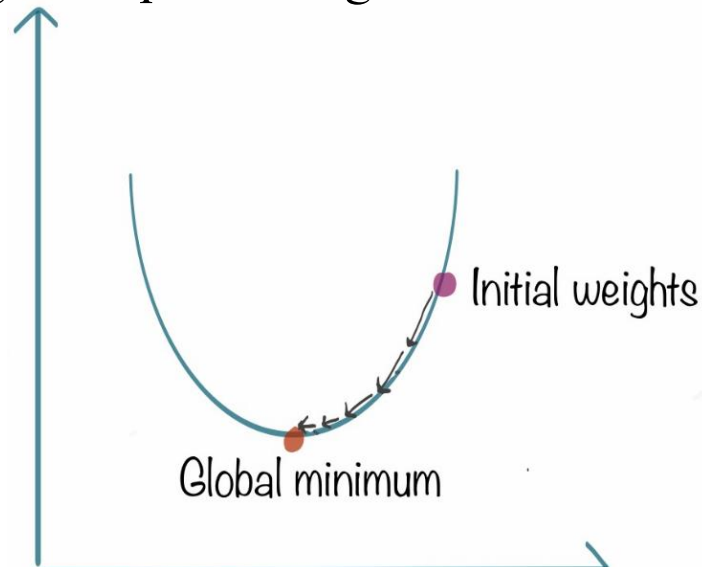


2000-12: Big Data, GPU, Deep Learning, NLP

- Google is popular (Big File, Google File Systems, Page Rank Algorithm etc.)
- Internet boom
- Shifted emphasis to big data – statistical techniques
- Graphical Processing Units (GPUs) inventions
- Spectacular results in Deep Neural Networks
- Specific goal oriented research expectations (Speech, Text, Image, Video etc)
- IBM's Watson beat the Jeopardy champions.
 - Due to Advance Natural Language Processing (NLP) and Q&A.

2000 onward: Deep Learning

- For the Probabilities and Expectation equations
 - an optimization algorithm finds the value of the parameters (weights) that minimize the error when mapping inputs to outputs.
 - Optimization algorithms affect the accuracy and the speed of the training a deep learning model.

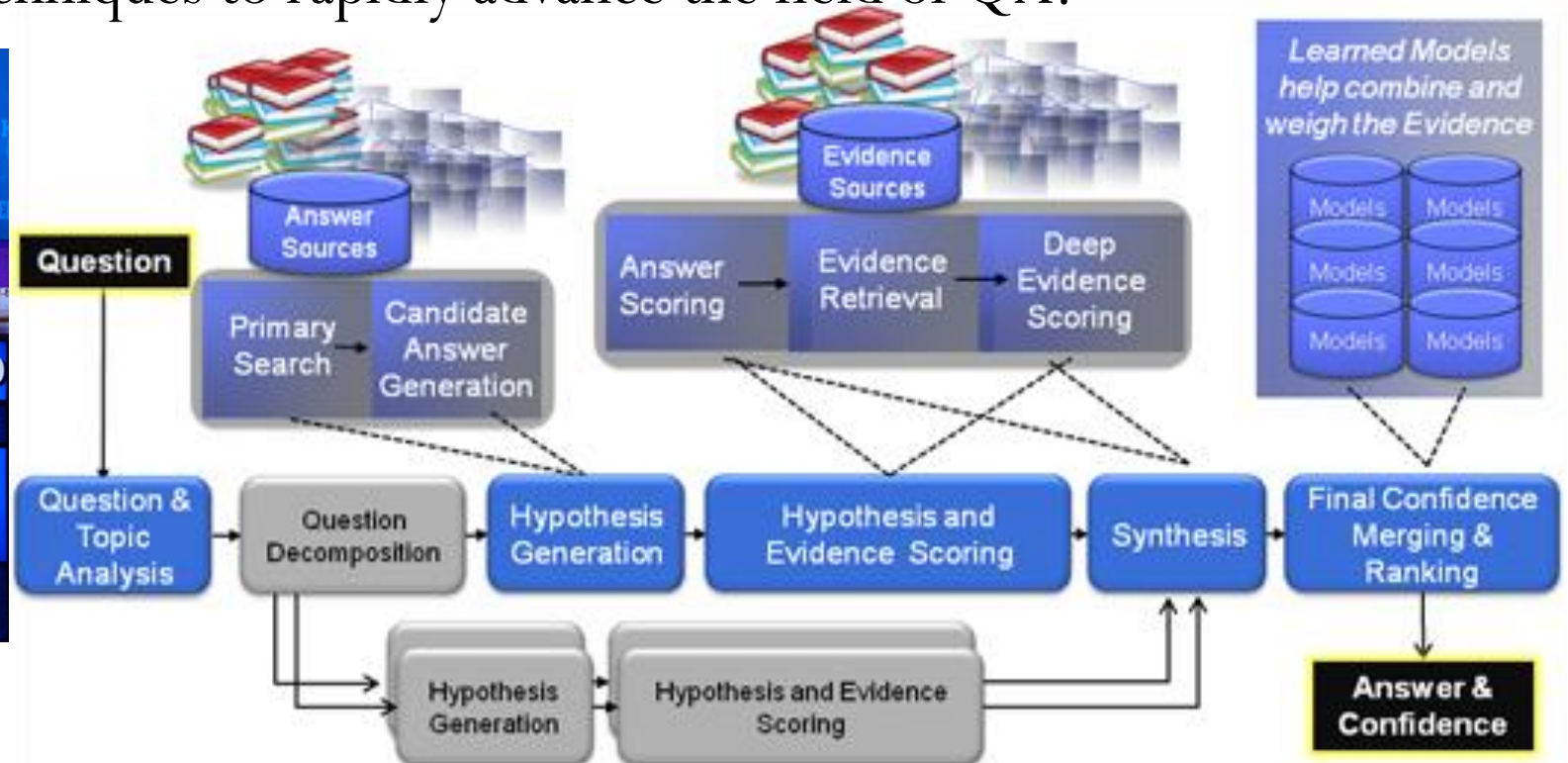


2011: IBM Watson beaten Human in Jeopardy

- Watson is performing at human expert levels in terms of precision, confidence, and speed at the Jeopardy quiz show.
- DeepQA architecture for combining, deploying, evaluating, and advancing a wide range of algorithmic techniques to rapidly advance the field of QA.



https://researcher.watson.ibm.com/researcher/view_group_subpage.php?id=2159

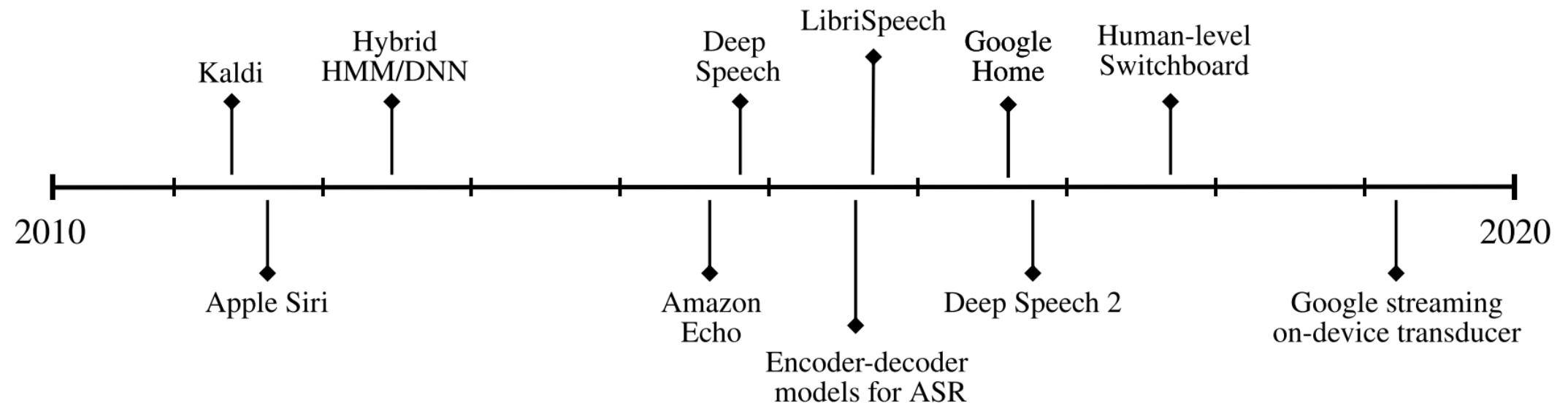


2010 onward: Natural Language - Speech

- Amazon Alexa
- Apple Siri
- Microsoft Cortana
- Google translate
- Voice recognition on Google Search, Whatsapp, etc.

Open-source and speech recognition software like Kaldi, and larger benchmarks like LibriSpeech are widely used.

ChaptGPT (2023)



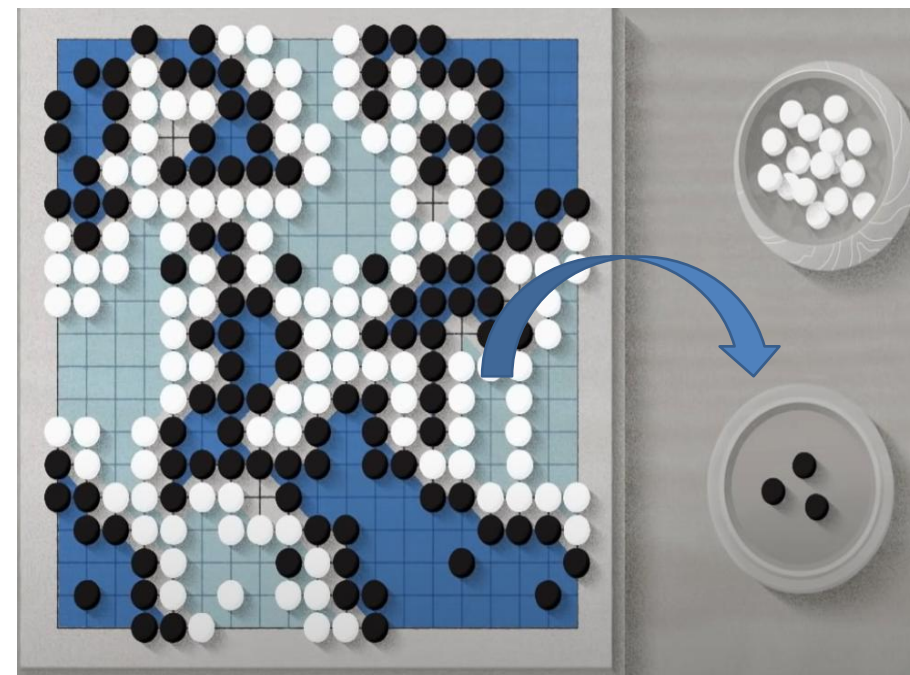
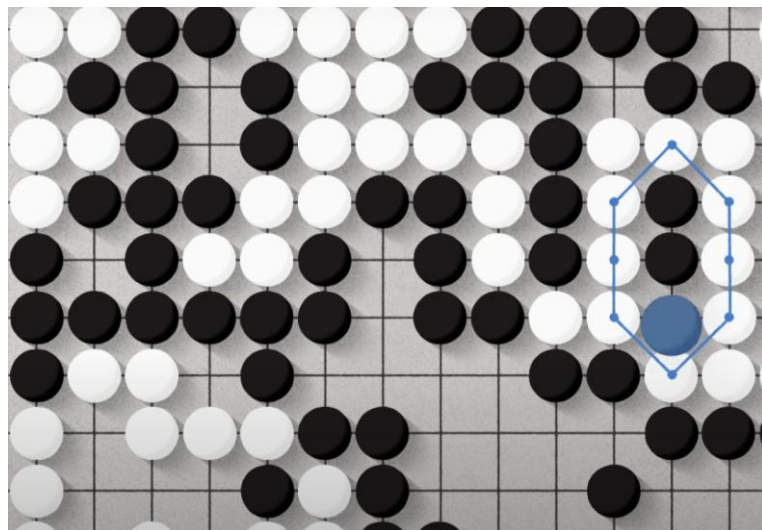
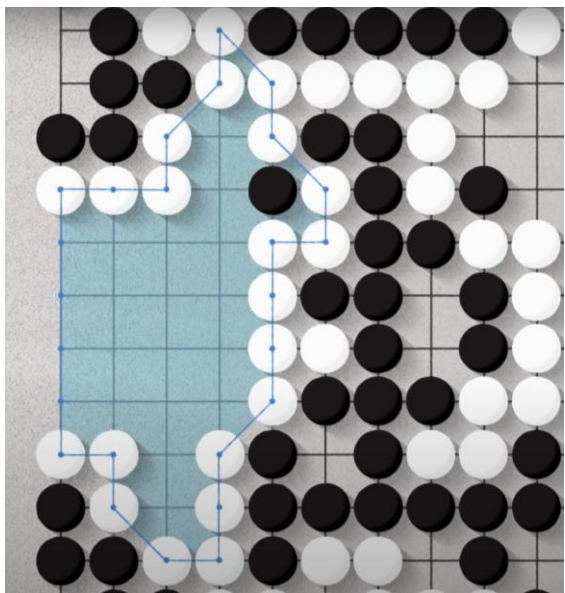
Hannun, Awni. "The history of speech recognition to the year 2030." *arXiv preprint arXiv:2108.00084* (2021).

2012 onward: AI Spring

- Private funding companies: Microsoft, Google, Facebook, IBM, AWS etc
- Fast development of computational power
- Large volume of data generation
- Availability of Big data: Growth of data
- Better data science, data mining, deep learning, and machine learning
- Better automated gaming due to Reinforcement Learning
- Better technology and huge computational power
- Cloud based Shared resources (Software and Hardware)
- Better algorithms, Self-supervised Learning, Open source software development

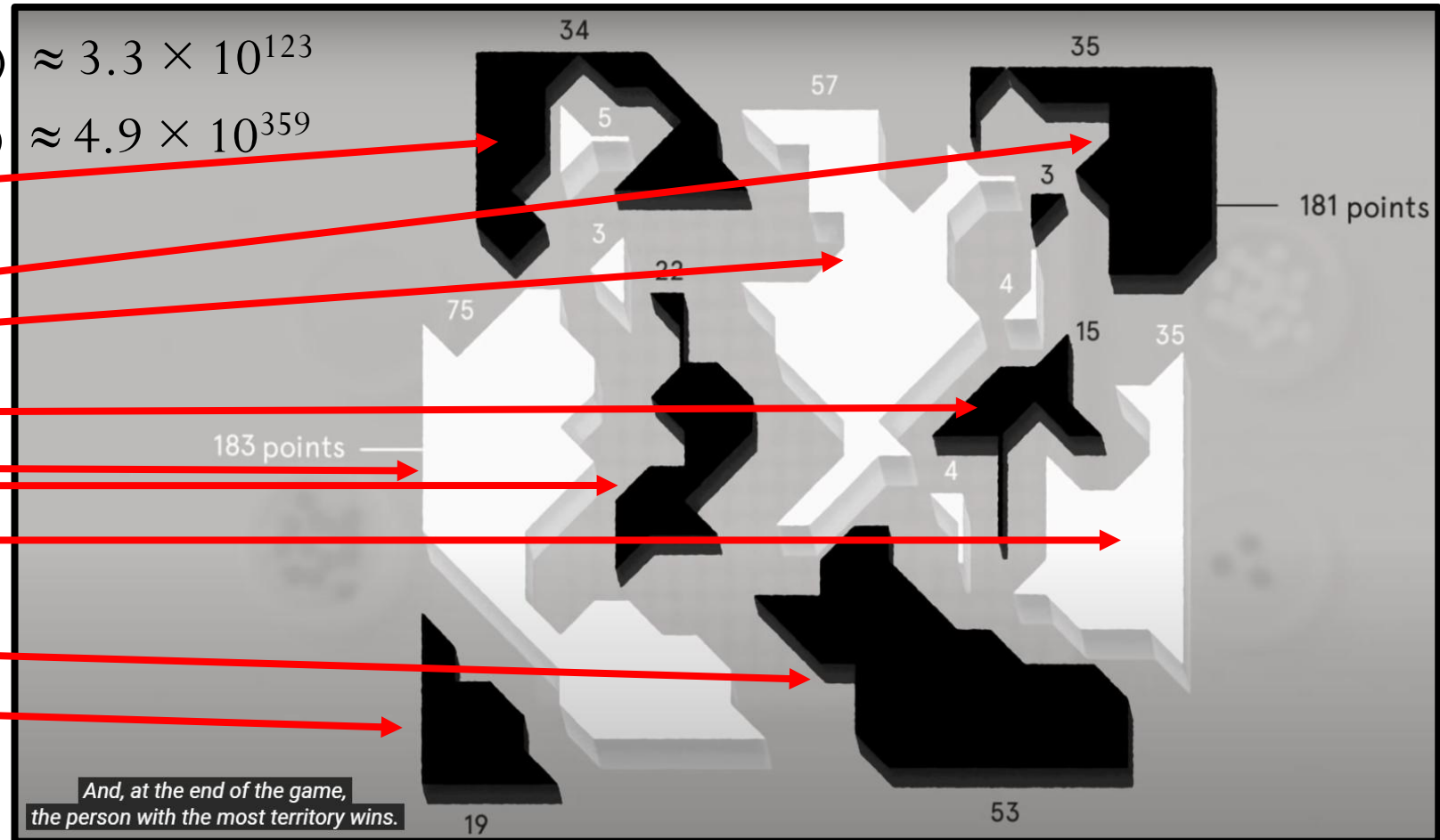
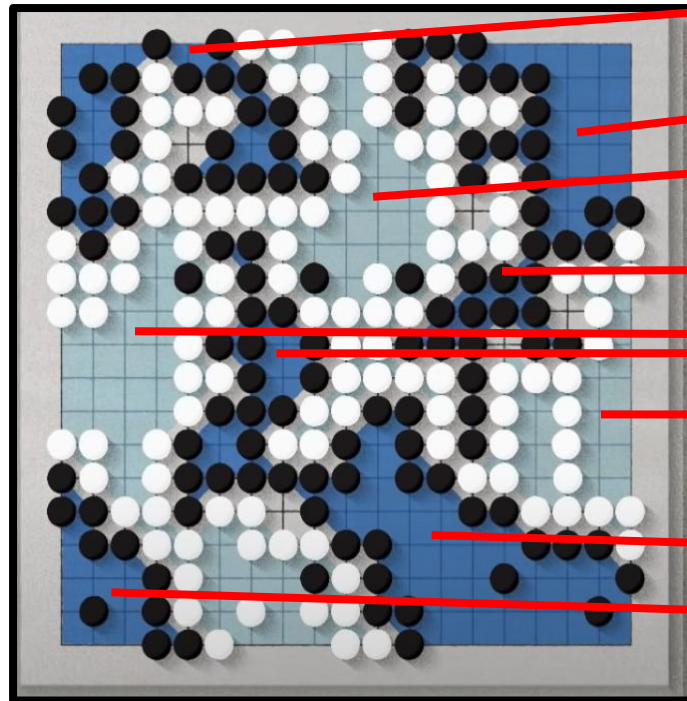
2016: Alpha Go

- Make territories
- Capture opponent pieces
- Largest territory



2016: Alpha Go

- Large games, search tree containing approximately b^d possible sequences of moves
 - exhaustive search is infeasible because
 - Chess ($b \approx 35$, $d \approx 80$) $\approx 3.3 \times 10^{123}$
 - Go ($b \approx 250$, $d \approx 150$) $\approx 4.9 \times 10^{359}$



2016: Alpha Go

- AlphaGo achieved a 99.8% winning rate against other Go programs, and defeated the human European Go champion by 5 games to 0.
- This is the first time that a computer program has defeated a human professional player in the full-sized game of Go.



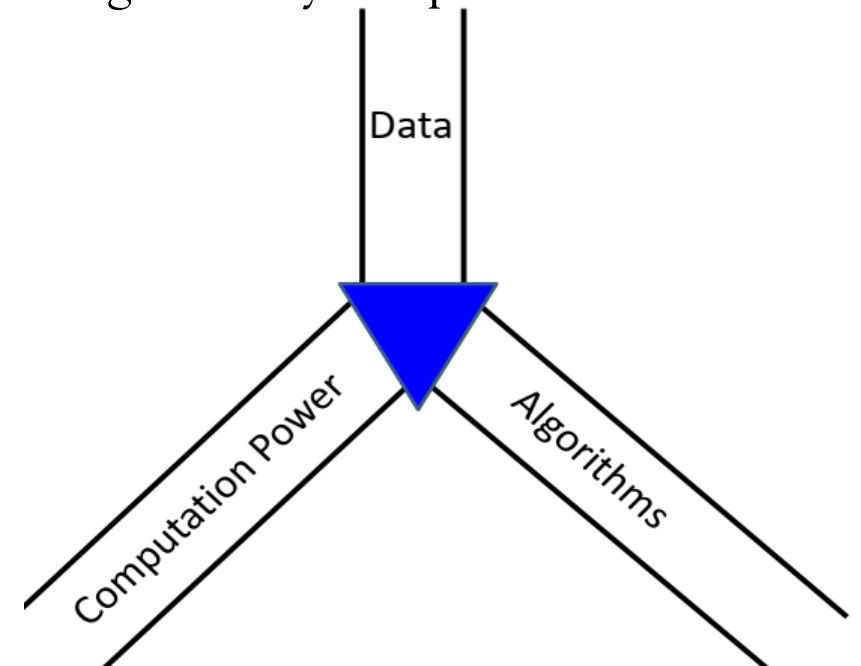
Improvement resulted better AI

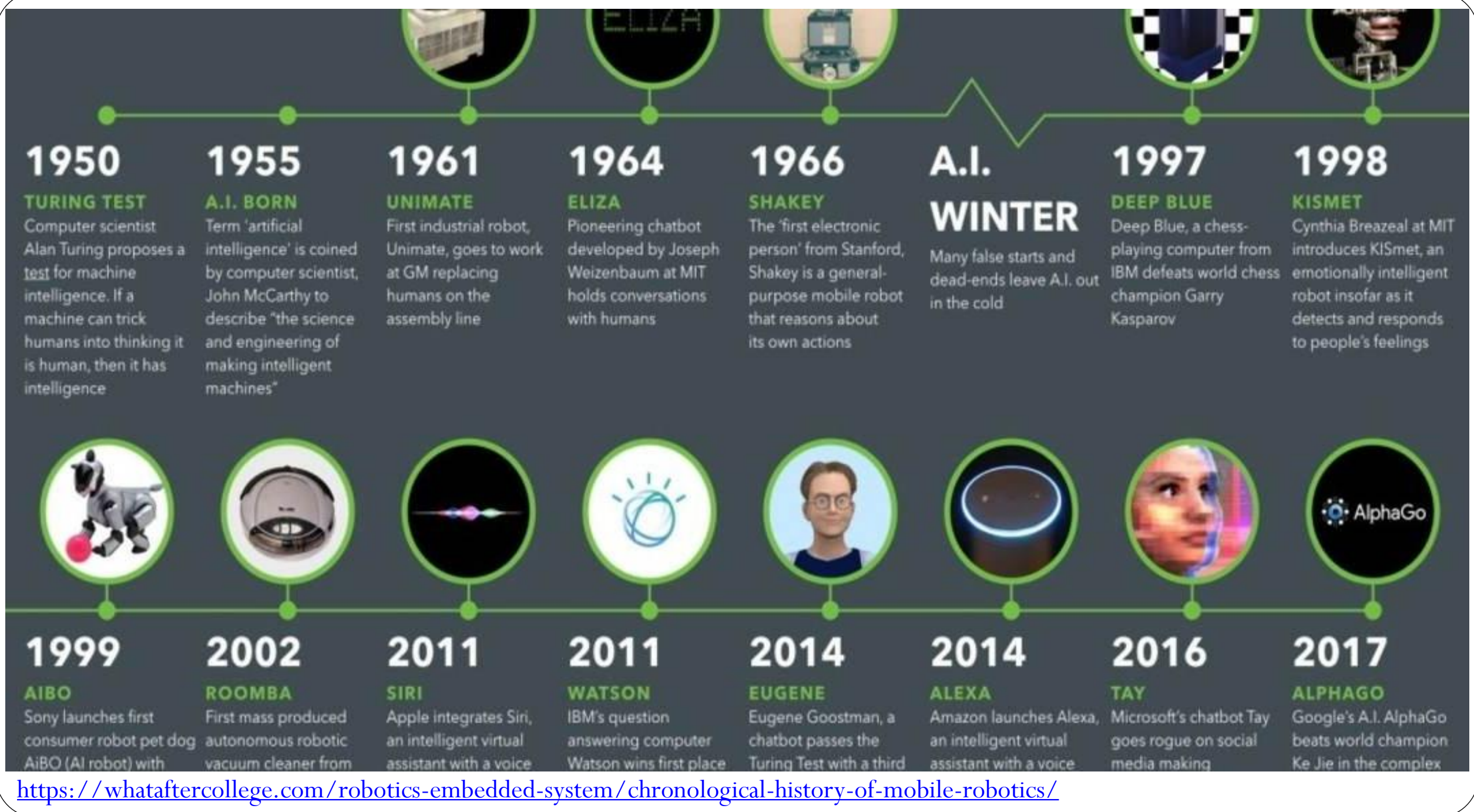
Problems

- sharp minima (leading to overfitting), non-robustness
 - analysis corresponds too closely or exactly to a particular set of data
 - therefore, fail to fit to additional data or predict future observations reliably
- numerical instability (vanishing/exploding gradients)
 - generally desirable property of numerical algorithms for solving ordinary and partial differential equations by discrete approximation

Why does AI work so well because

- high capacity (susceptible to overfitting)
- Better and more training Data
- Better, large, scalable, parallel Computational Power
- Better, parallel and distributed Algorithms
- Slowly but steadily increasing emphasis on
 - Explainability and Interpretability
 - Theoretical justifications





1950

TURING TEST

Computer scientist Alan Turing proposes a test for machine intelligence. If a machine can trick humans into thinking it is human, then it has intelligence

1955

A.I. BORN

Term 'artificial intelligence' is coined by computer scientist, John McCarthy to describe "the science and engineering of making intelligent machines"

1961

UNIMATE

First industrial robot, Unimate, goes to work at GM replacing humans on the assembly line

1964

ELIZA

Pioneering chatbot developed by Joseph Weizenbaum at MIT holds conversations with humans

1966

SHAKY

The 'first electronic person' from Stanford, Shakey is a general-purpose mobile robot that reasons about its own actions

**A.I.
WINTER**

Many false starts and dead-ends leave A.I. out in the cold

1997

DEEP BLUE

Deep Blue, a chess-playing computer from IBM defeats world chess champion Garry Kasparov

1998

KISMET

Cynthia Breazeal at MIT introduces KISmet, an emotionally intelligent robot insofar as it detects and responds to people's feelings

1999

AIBO

Sony launches first consumer robot pet dog AiBO (AI robot) with

2002

ROOMBA

First mass produced autonomous robotic vacuum cleaner from

2011

SIRI

Apple integrates Siri, an intelligent virtual assistant with a voice

2011

WATSON

IBM's question answering computer Watson wins first place

2014

EUGENE

Eugene Goostman, a chatbot passes the Turing Test with a third

2014

ALEXA

Amazon launches Alexa, an intelligent virtual assistant with a voice

2016

TAY

Microsoft's chatbot Tay goes rogue on social media making

2017

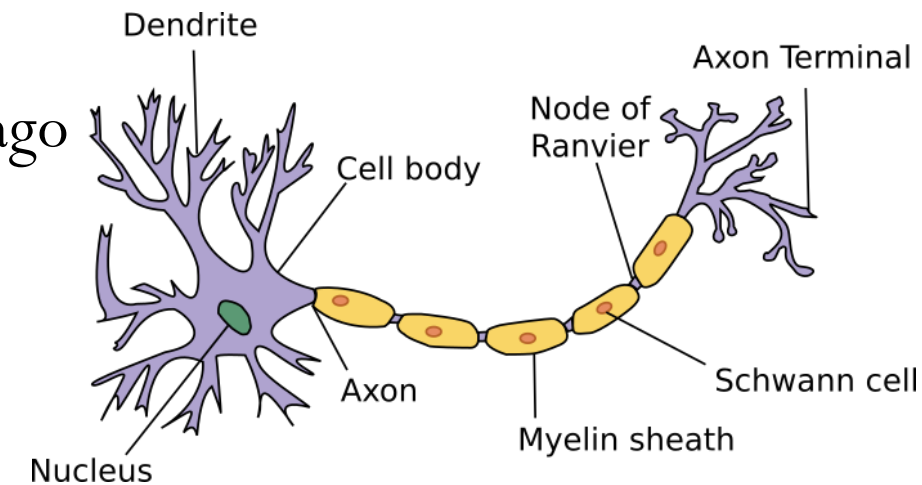
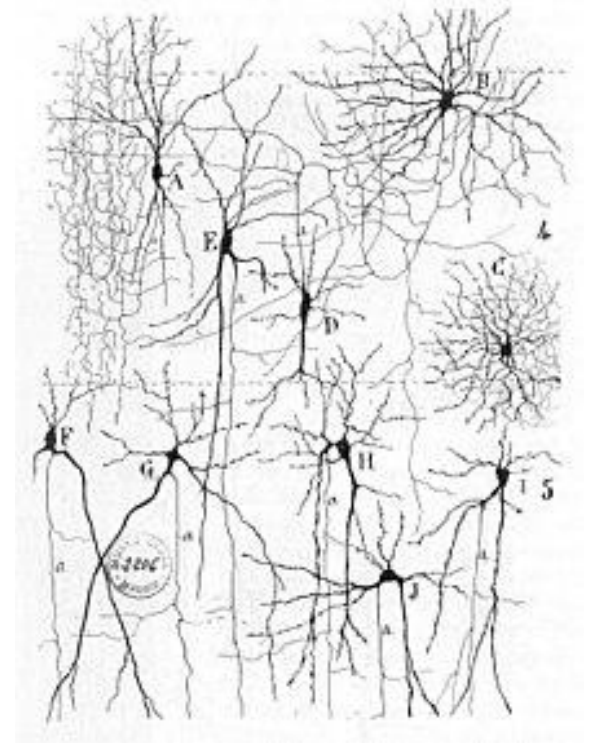
ALPHAGO

Google's A.I. AlphaGo beats world champion Ke Jie in the complex

History of Artificial Neural Network (ANN)

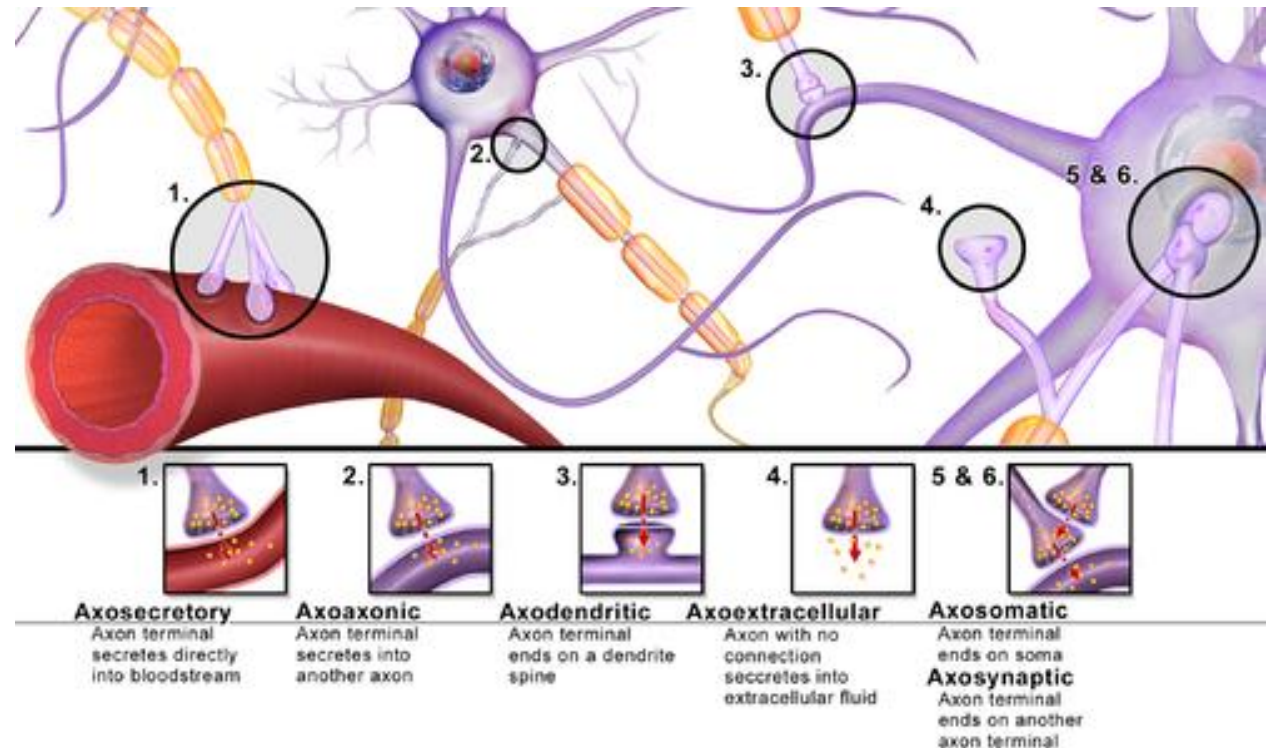
Neural Network (Anatomy)

- **1870s:** Reticular Theory “nervous system and brain is a single continuous network, all nerve cells in the nervous system constituted a continuous, interconnected network”
- **1888:** Neurons “fundamental units of the brain and nervous system, each nerve cell is an independent entity and nerve synapses transfer nerve impulses from one cell to another”
- **1906:** Nobel prize to both Camillo Golgi and Santiago Ramón y Cajal "in recognition of their work on the structure of the nervous system."



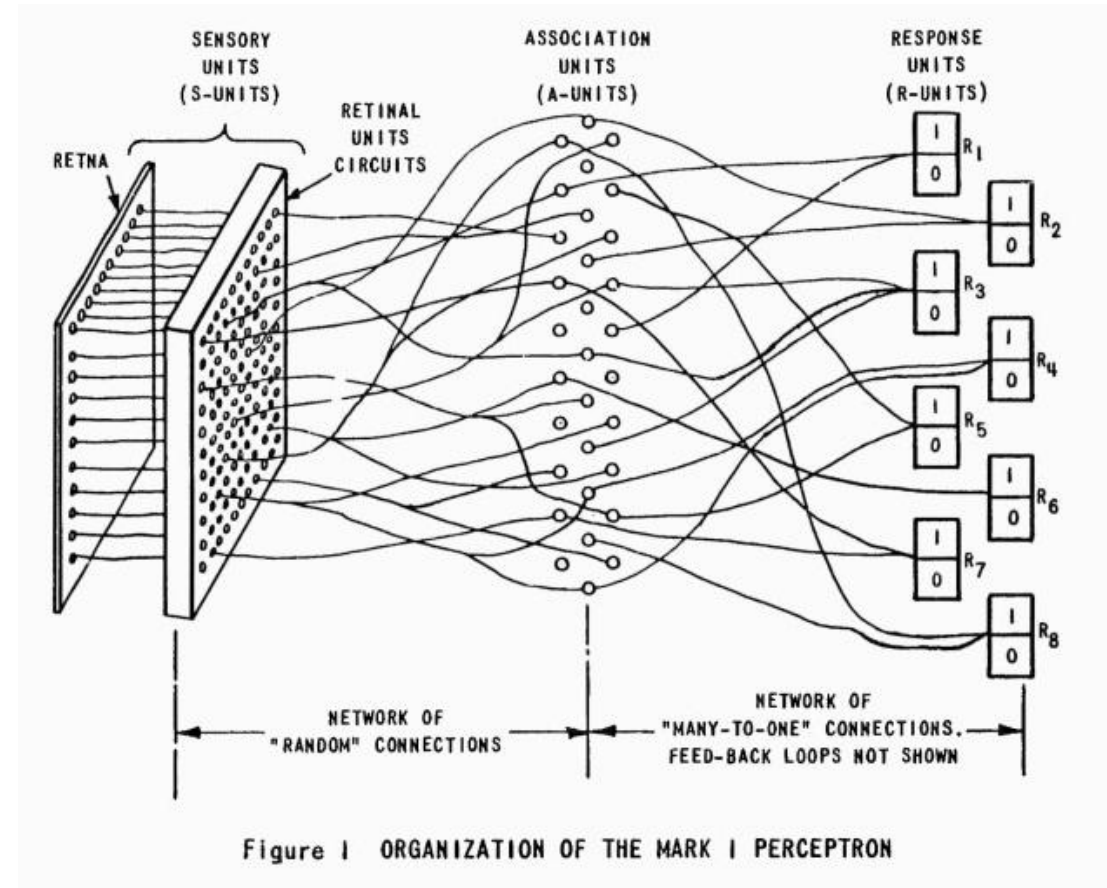
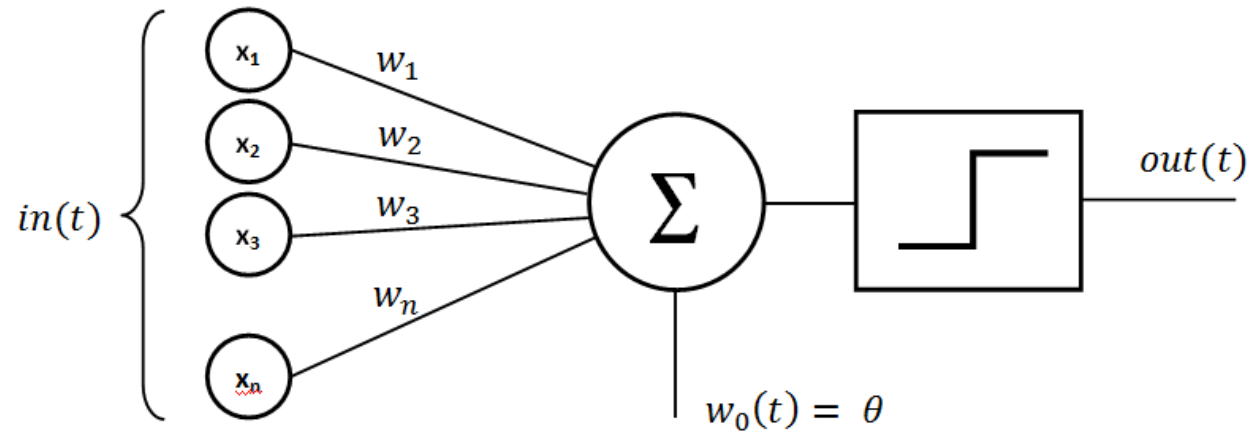
Neural Network (Anatomy)

- Synapse is a structure that permits a neuron (or nerve cell) to pass an electrical or chemical signal to another neuron or to the target effector cell.
- Synapses are essential to the transmission of nervous impulses from one neuron to another.
- Different types of synapses



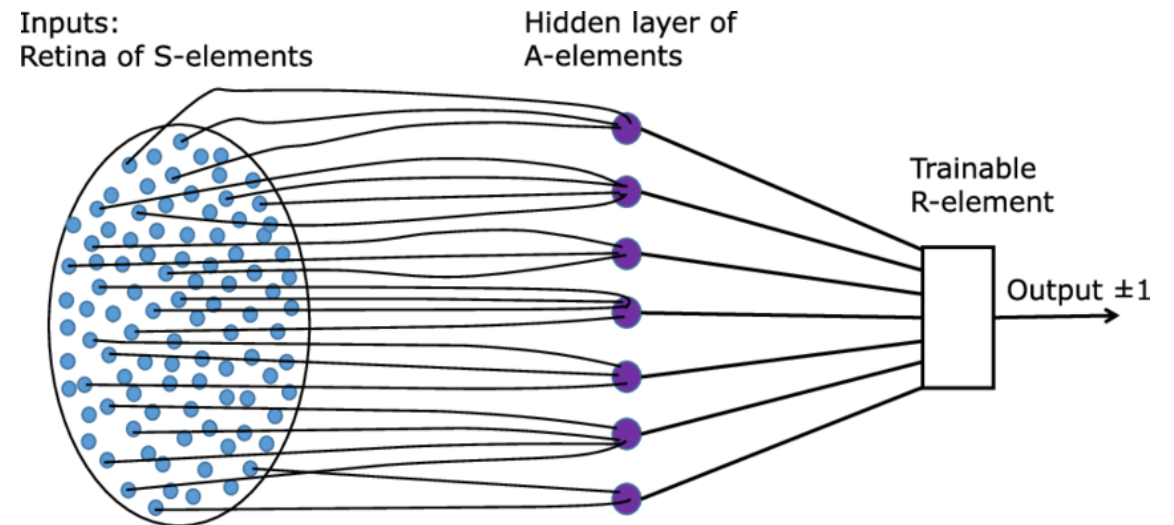
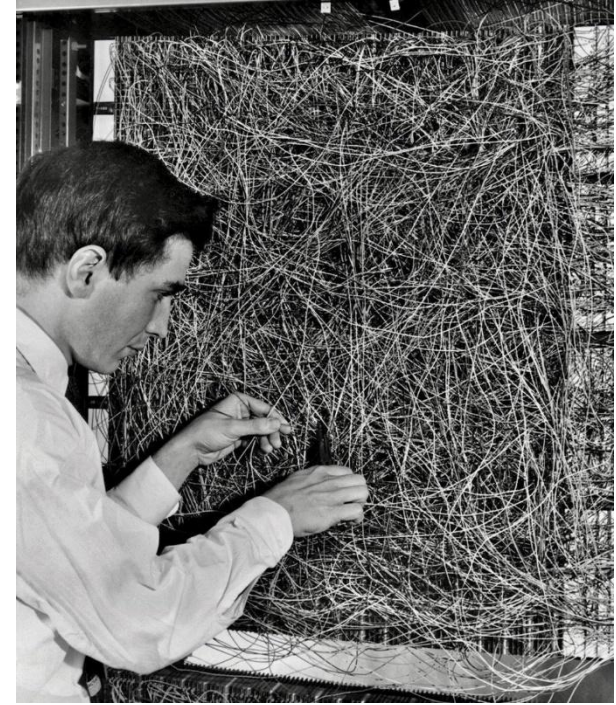
Perceptron

- 1943 Perceptron was invented by **McCulloch and Pitts**
- 1958 **Mark I Perceptron** hardware developed and constructed by **Frank Rosenblatt**

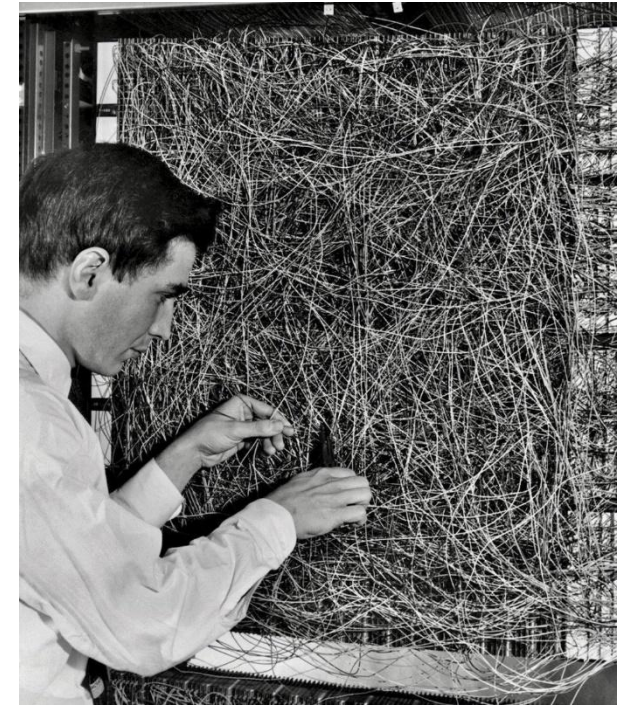
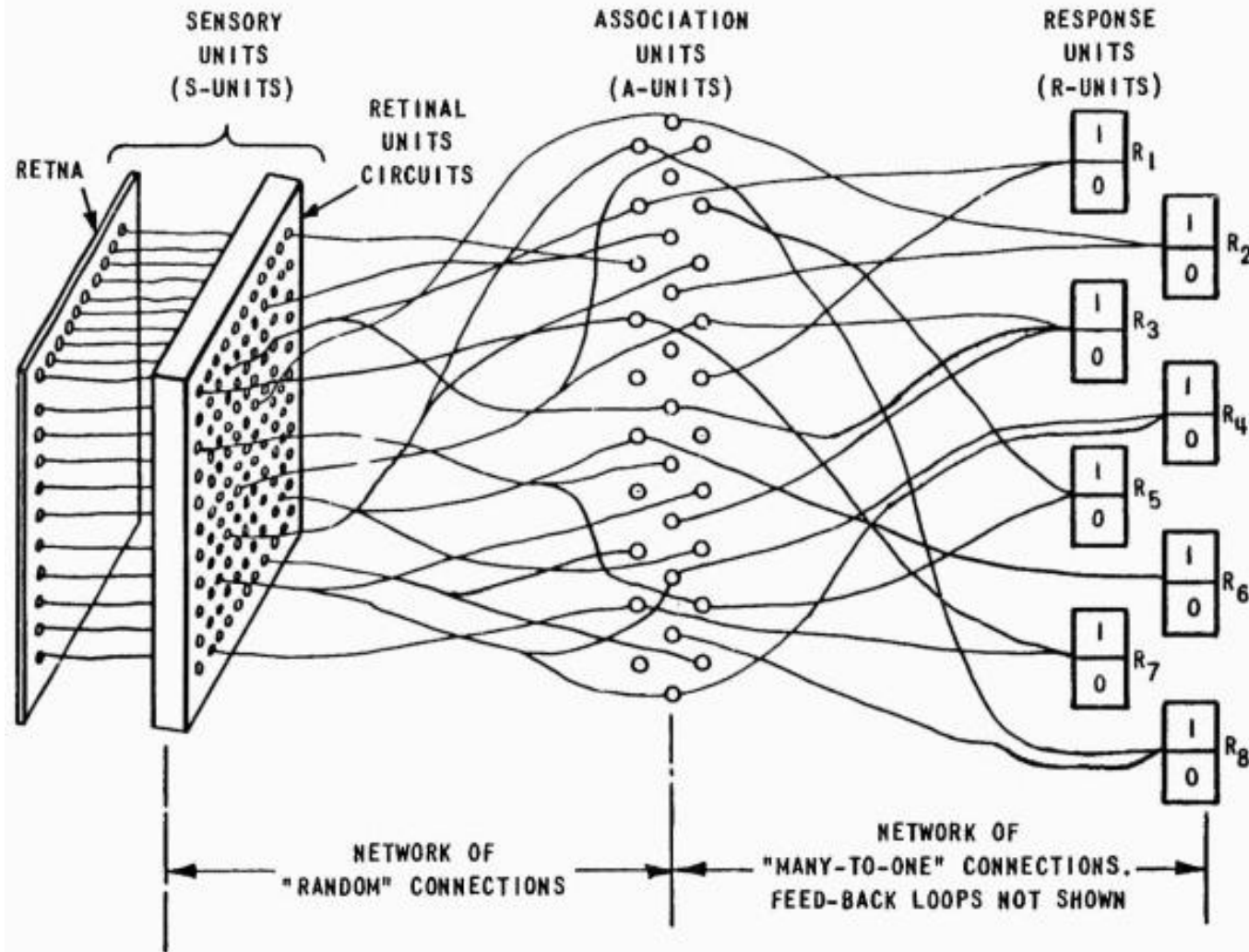


Perceptron

- 1958 Frank Rosenblatt development and hardware construction of the “**Mark I Perceptron**”
 - the first computer that could learn new skills by trial and error, using a type of neural network that simulates human thought processes.
 - at the Cornell Aeronautical Laboratory funded by the United States Office of Naval Research.

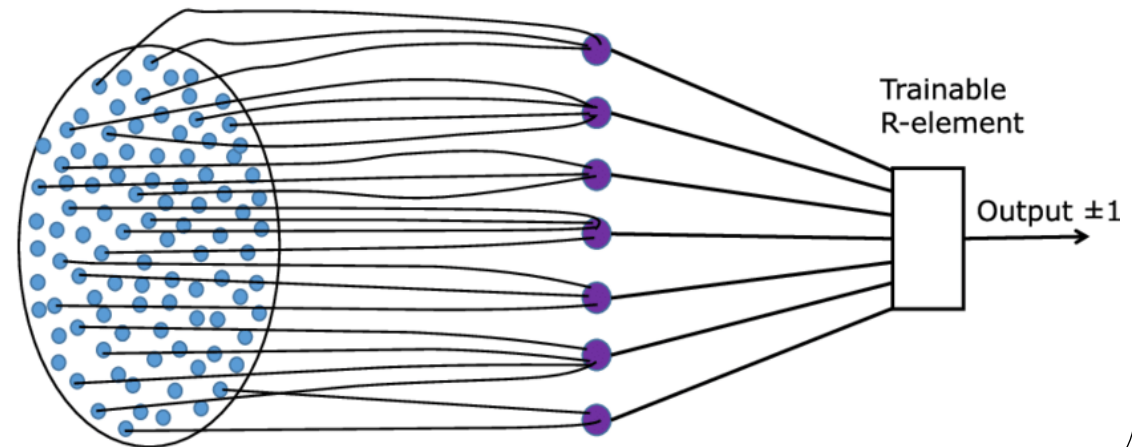


Perceptron



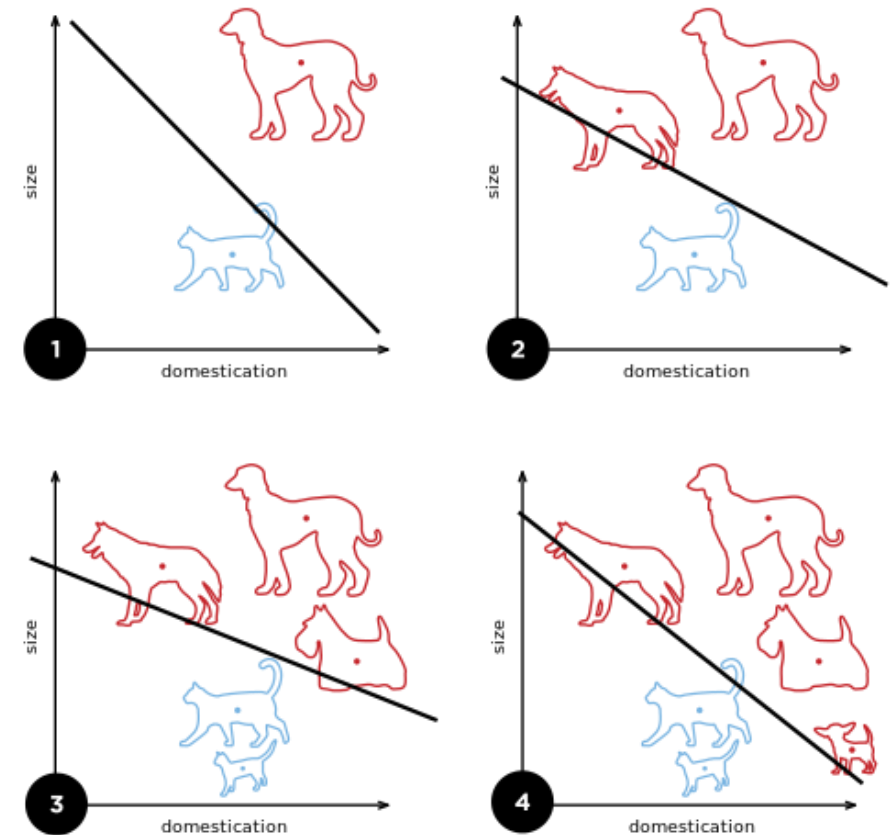
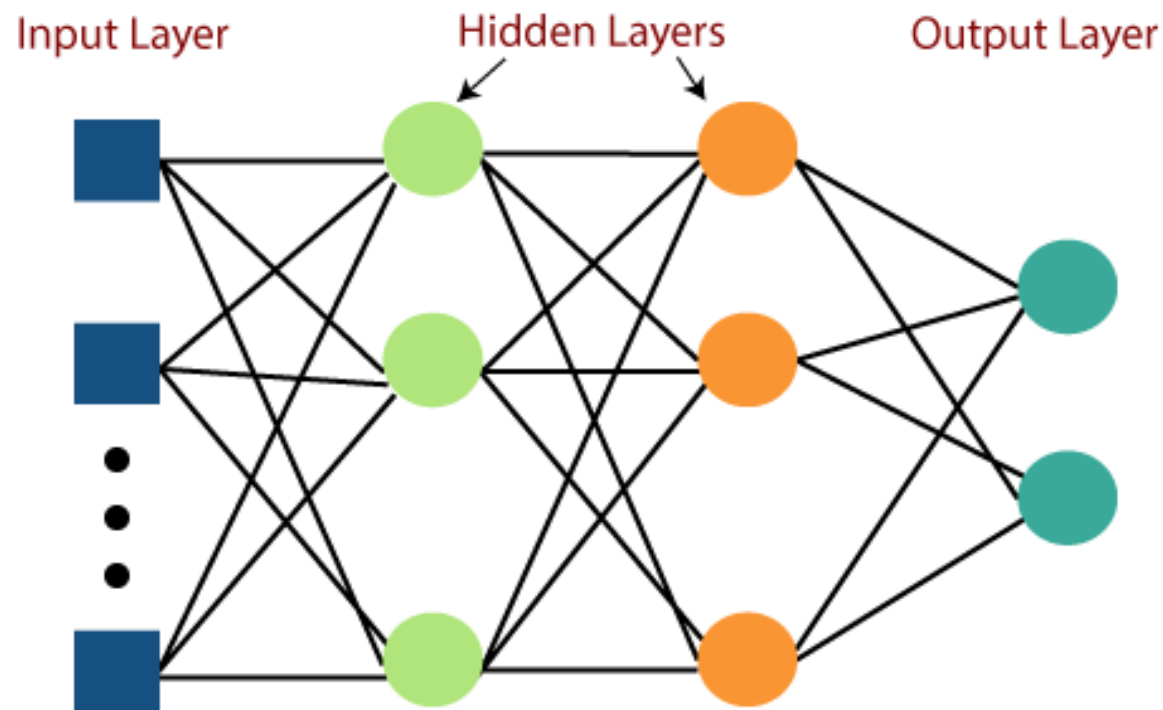
Inputs:
Retina of S-elements

Hidden layer of
A-elements



Multi-layer Perceptron

- **1960s** Limitation of Perceptron: Failed to classify basic problems
- Ivakhnenko et. al. introduced Multilayer Perceptrons

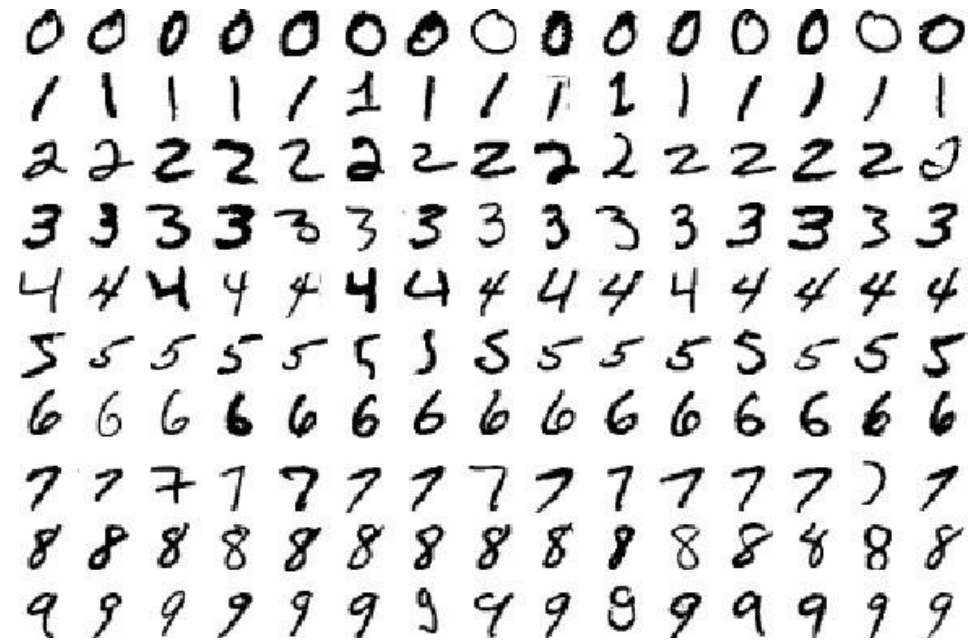


Artificial Neural Network and Backpropagation

- 1982: Werbos introduced Backpropagation first used it in the context of Artificial Neural Networks
- 1986: Rumelhart et. al. popularized the Backpropagation
- Use of Cauchy discovered theories of Gradient Descent on Convex function for Optimization
- 1989: Universal Approximation Theorem “A multilayered network of neurons with a single hidden layer can be used to approximate any continuous function to any desired precision”

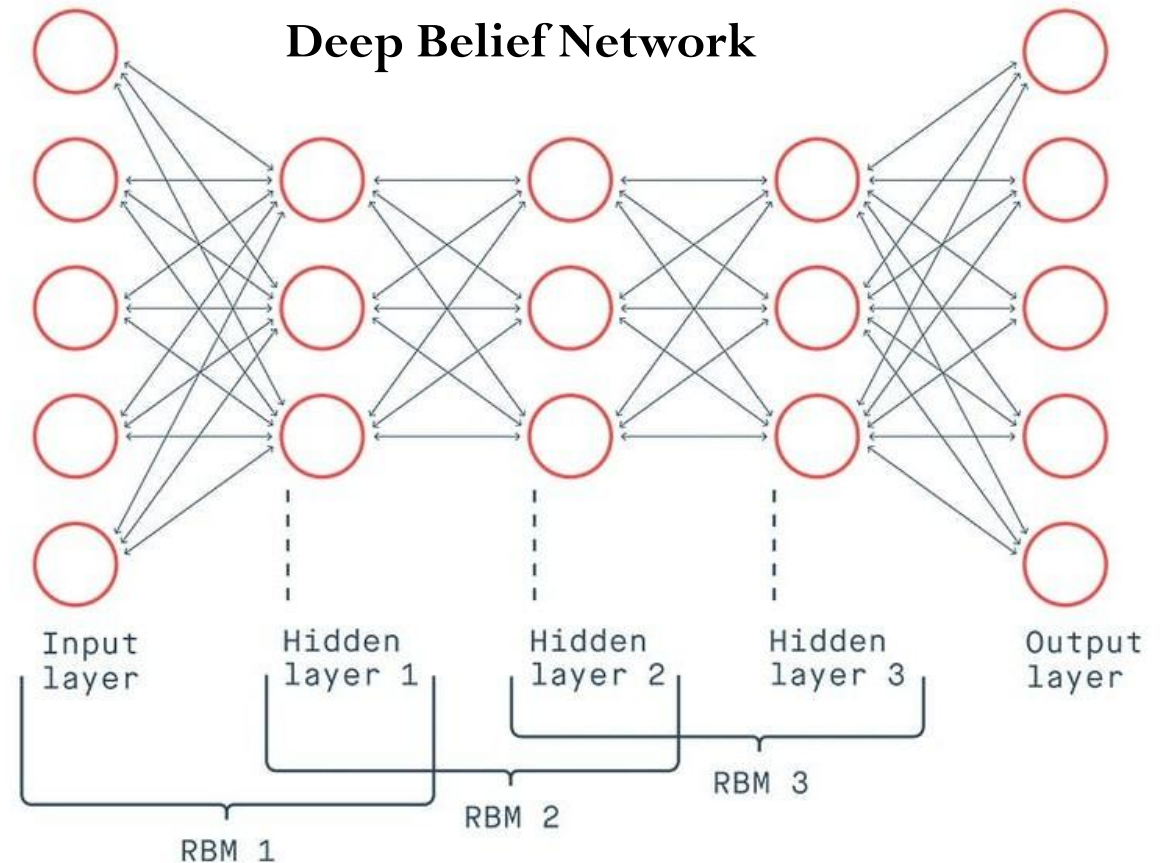
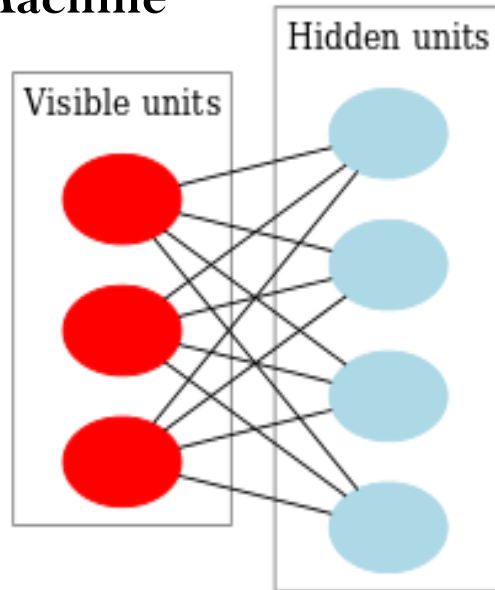
Deep Learning

- Unsupervised Pre-Training
 - 1991-1993: J. Schmidhuber “Very Deep Learner”
 - 2006: Hinton and Salakhutdinov
- 2009: Handwriting Recognition
 - MNIST dataset
- 2010: Speech Recognition
 - Dahl et. al. achieved error reduction of 16.0% to 23.2% over previous works
- Traffic Sign Recognition Competition
 - D. C. Ciresan et. al. got only 0.56% error rate



Restricted Boltzmann Machine and Deep Belief Network

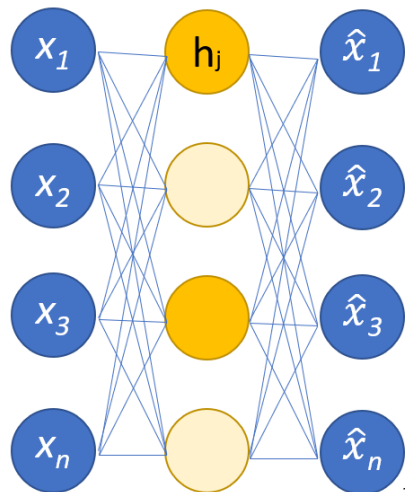
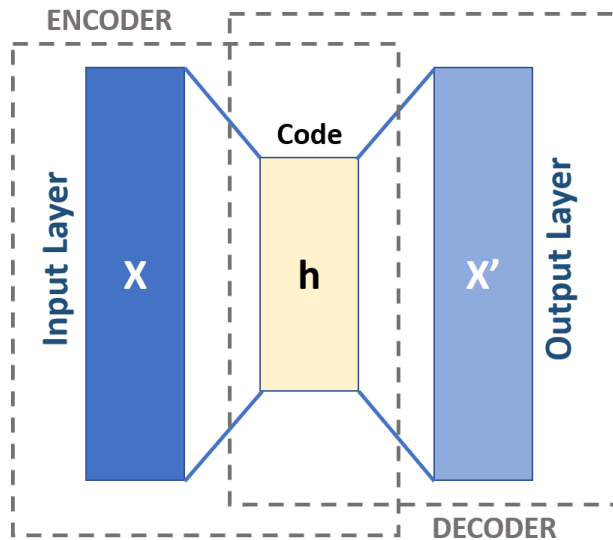
Restricted Boltzmann Machine



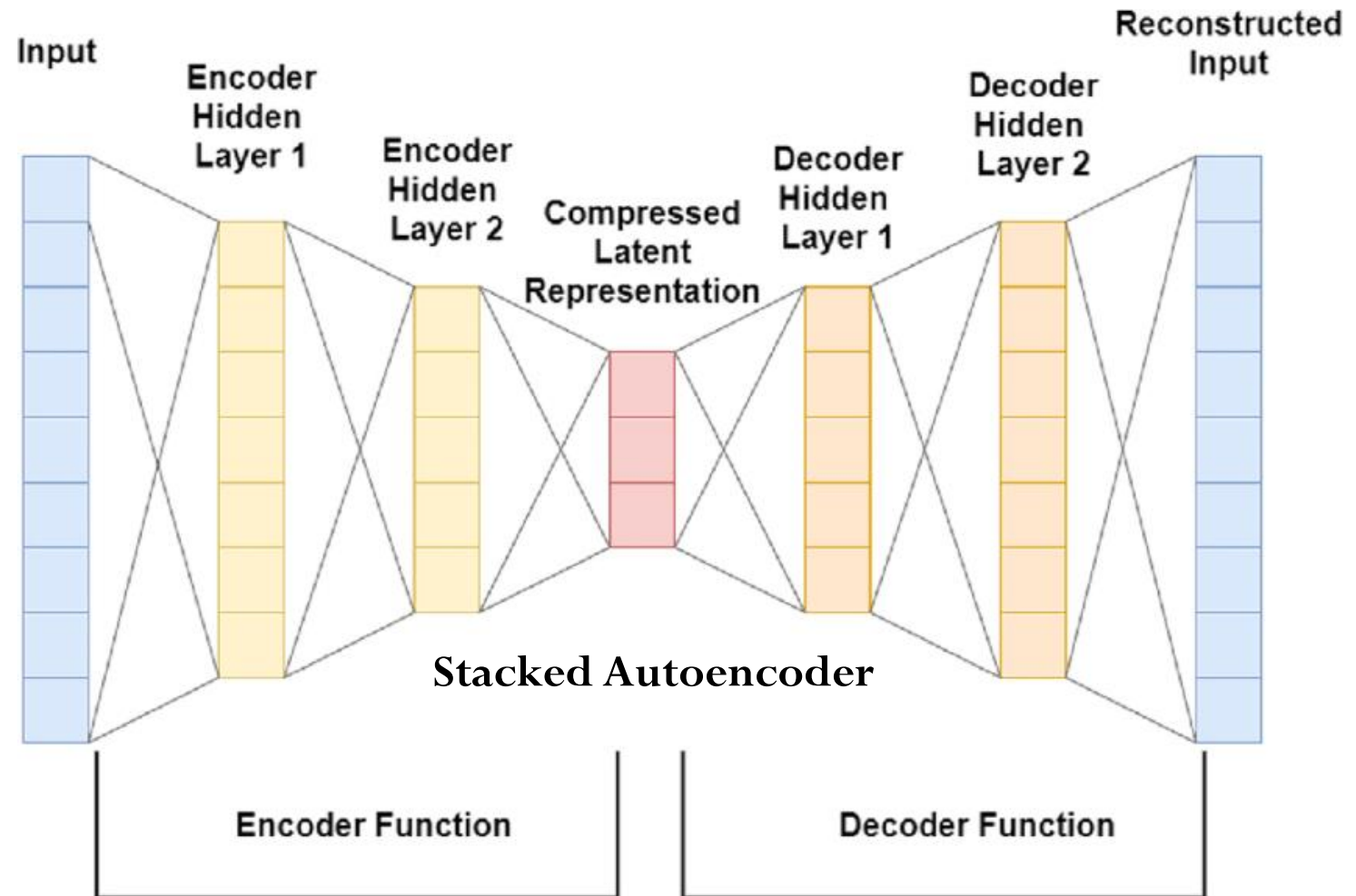
Ruslan Salakhutdinov, Andriy Mnih, and Geoffrey Hinton. "Restricted Boltzmann machines for collaborative filtering." Proceedings of the 24th International Conference on Machine learning. 2007.

Geoffrey E. Hinton, Simon Osindero, and Yee-Whye Teh. "A fast learning algorithm for deep belief nets." Neural computation 18.7 (2006): 1527-1554.

Autoencoder and Stacked Autoencoder



Autoencoder

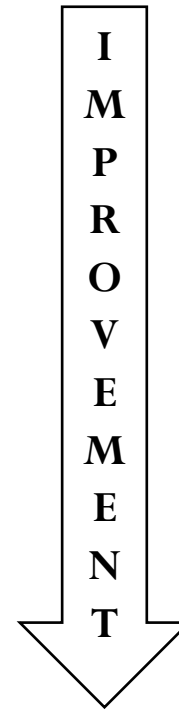


Pascal Vincent, et al. "Stacked denoising autoencoders: Learning useful representations in a deep network with a local denoising criterion." Journal of Machine Learning Research 11.12 (2010).

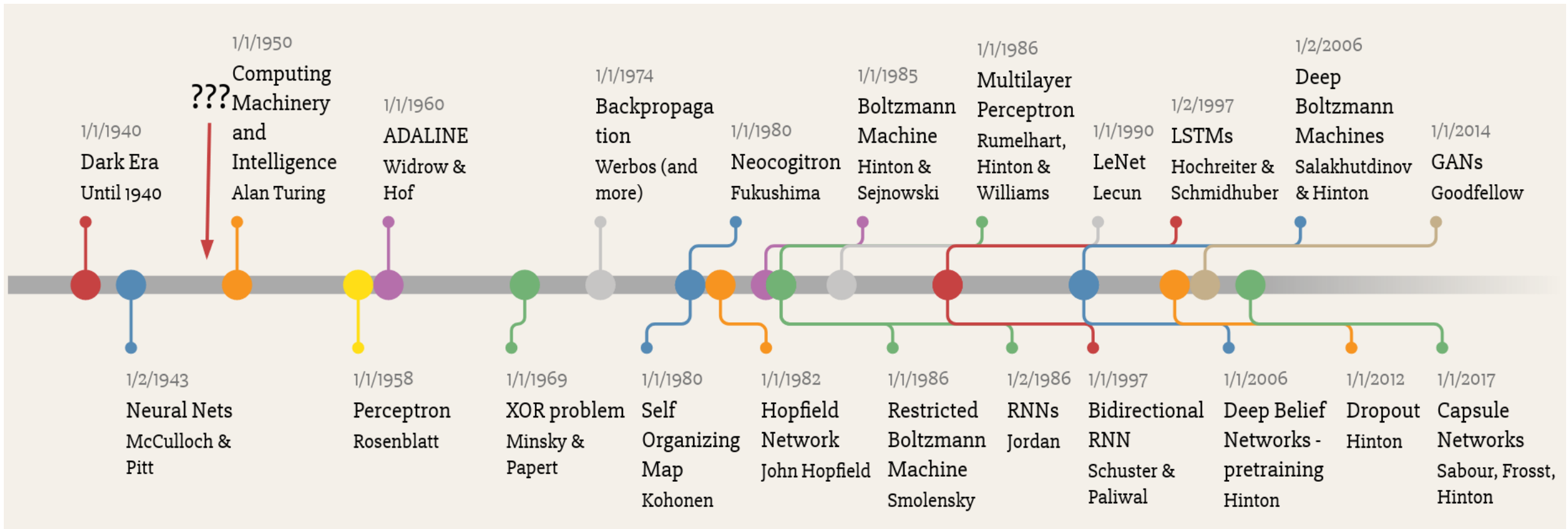
Visual Recognition Challenges

- ImageNet challenge successful 2012-2016

Network	Error	Layers
• AlexNet	16.0%	8
• ZFNet	11.2%	8
• VGGNet	7.3%	19
• GoogLeNet	6.7%	22
• MS ResNet	3.6%	152!!



Artificial Neural Network (ANN) Timeline



Deep Learning for the Masses (... and The Semantic Layer), Deep Learning timeline by Favio Vazquez

<https://www.kdnuggets.com/2018/11/deep-learning-masses-semantic-layer.html>

Transformer and Chat GPT

- Dr. Ashish Vaswani was a research scientist in the Google Brain team.
- Lead-author of the paper “Attention Is All You Need,” which introduced the Transformer model, in 2017.
- Transformer architecture has revolutionized natural language processing (NLP).
- Transformer architecture on recurrent neural networks (RNNs) and Attention.
- GPT-1, 2018, NLP benchmarks.
- OpenAI has released several updated versions of the GPT model.
- The ChatGPT model is based on the GPT based on the Transformer model.

<https://viterbischool.usc.edu/news/2023/03/attention-is-all-you-need-usc-alumni-paved-path-for-chatgpt/>
https://medium.com/@kishore_chandra/relation-between-chat-gpt-and-ashish-vaswani-3a4b07528a80

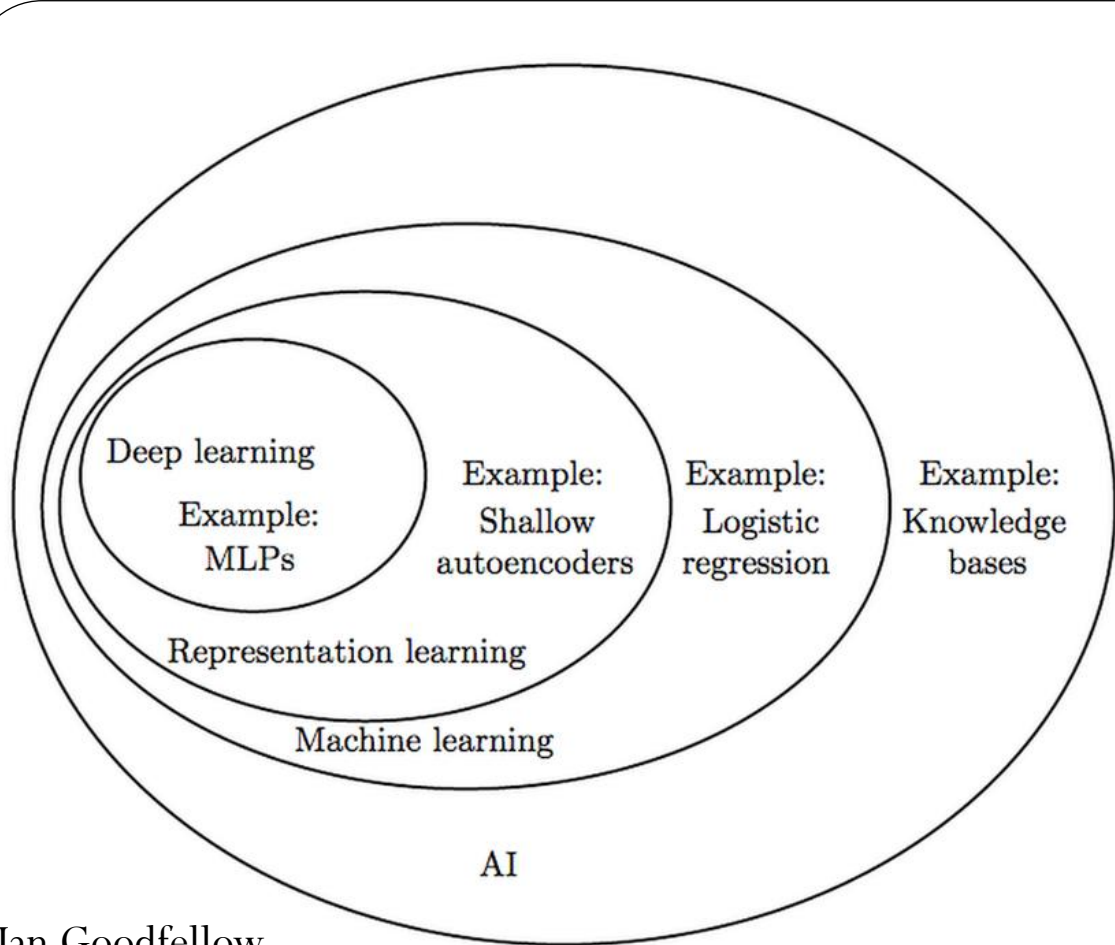
Summary

Artificial Intelligence

- Founded in 1956 (Dartmouth Research Project on Artificial Intelligence).
- Ancient History shows that AI was an active field of studies since ancient times.
- Simulating the brain, modeling human problem solving, formal logic, large databases of knowledge and imitating human behavior
- Search, Optimization, Formal logic,
- Artificial Neural Networks, Deep learning,
- Classifiers, Reasoning, Knowledge Representation,
- Planning, Learning, Robotics,
- Natural Language Processing, Computational Social Scienc, etc.

Artificial Intelligence

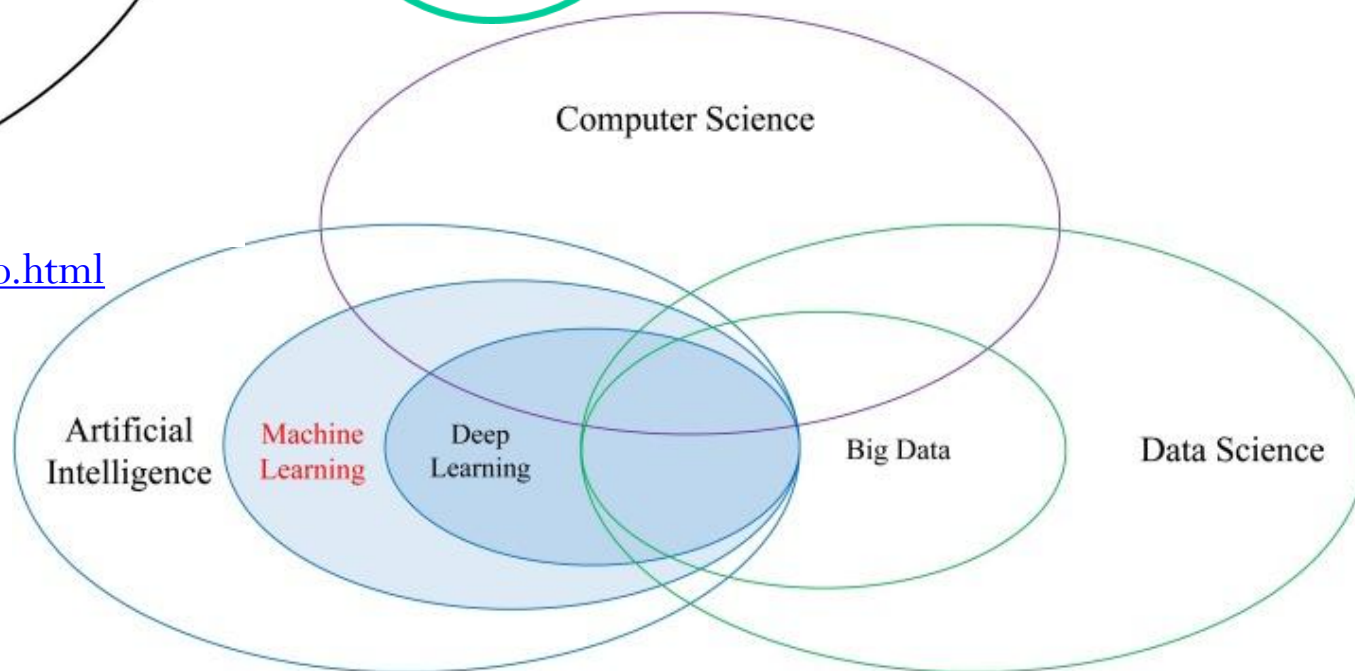
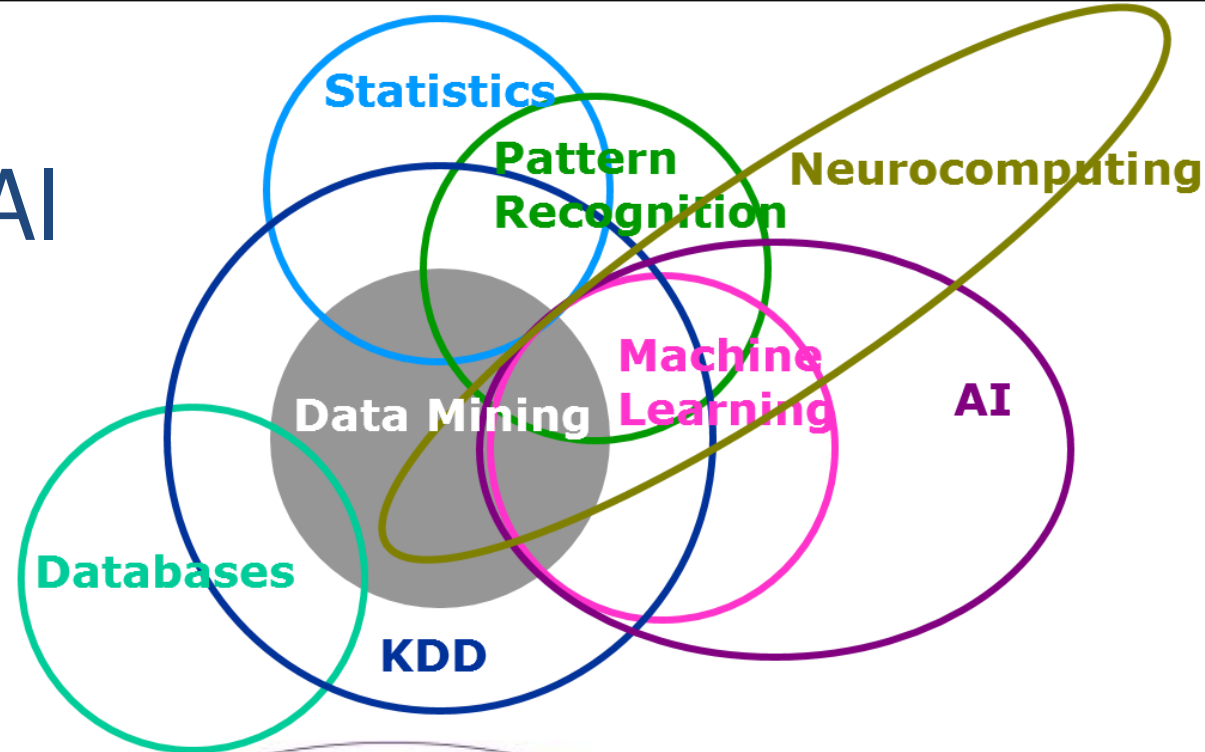
- (1837) Analytical Engine, (1925) Houdina Radio Control,
- (1946-1960) ENIAC heralds the dawn of Computing,
- (1946-1960) Turing Test, (1959) Machine Learning,
- (1964) Eliza Chatbot, (1966) Shakey general purpose mobile robot,
- (1960-74) Expert Systems,
- AI Winter (1974-80), AI Spring (1980-87), AI Winter (1987-93),
- (1993) Apriori algorithm: Association Rule Mining,
- (1997) Deep Blue ends Human Supremacy in Chess,
- (1998) Deep Space, (2005) Driverless Cars,
- (2011) IBM Watson beaten Human in Jeopardy,
- (2016) Alpha Go



Ian Goodfellow

<https://www.deeplearningbook.org/contents/intro.html>

AI



References

- Stuart Russel, and Peter Norvig. "Artificial intelligence: A modern approach. Third edit." Upper Saddle River, New Jersey 7458 (2015). <http://aima.cs.berkeley.edu/>
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- Images are from several sources e.g. movies, TV serials, internet, miscellaneous links, slides, blogs, etc.
- Subbarao Kambhampati, Arizona State University, <https://rakaposhi.eas.asu.edu/cse471/>
- Artificial Intelligence, Prof. Mausam <https://www.cse.iitd.ac.in/~mausam/courses/col333/autumn2019/>

תודה רבה

Hebrew

Ευχαριστώ

Greek

Спасибо

Russian

Danke

German

Merci

French

धन्यवादः

Sanskrit

நன்றி

Tamil

شكراً

Arabic

ಧನ್ಯವಾದಗಳು

Kannada

Thank You

English

നന്നി

Malayalam

Grazie

Italian

ధన్యవాదాలు

Telugu

આભાર

Gujarati

多謝

Traditional Chinese

Gracias

Spanish

ਧੰਨਵਾਦ

Punjabi

धन्यवाद

Hindi & Marathi

多谢

Simplified Chinese

<https://sites.google.com/site/animeshchaturvedi07>

Obrigado

Portuguese

ありがとうございました

Japanese

ขอบคุณ

Thai

감사합니다

Korean